

Advanced

Follow-along guide: Work with Tableau, Part 2

This document includes detailed instructions for how to perform the data visualizations described in the video “Work with Tableau, Part 2.”

The following guide points out areas of the video that may require adjustment. These reference guides can also serve as a set of usability reminders for you to recall when using Tableau in your future career.

Instructions

- Go to <https://public.tableau.com/s/>
- Since you’ve already [set up your Tableau Public profile](#), all you need to do is log in and select **Web Authoring** under **Create** in the navigation bar.
- Select the appropriate CSV file provided in the [instructions](#). The dataset you’ll use with this instructional video is: tableau_dataset.csv.
 - Please be aware that when you download the zip file folder provided, the computer automatically names that zip file folder with a long string of numbers and letters. You have to open that folder and then upload the individual files that are named correctly and match what's shown in the video.

(Note: Please allow several minutes for dataset upload.)

Before you can start designing visualizations, you’ll first need to upload your data. You’ll need to upload the specific dataset files to Tableau. Do not upload the entire .zip folder. When you download the zip folder from this page, your computer will automatically download a .zip file folder. The .zip folder is automatically named with a series of letters and numbers. Open that .zip folder, then save the individual dataset files. The two files are:

tableau_main_2009_to_2018.csv and tableau_dataset.csv. Once you can see the individual dataset files, proceed to upload your dataset for this video to Tableau Public.

Notice on the data source tab that you can see all of your column headers and Tableau icons that help you determine data types. In this case, you’ll see a calendar icon, globe icons, and pound signs.

tableau_dataset

tableau_dataset.csv



Need more data?

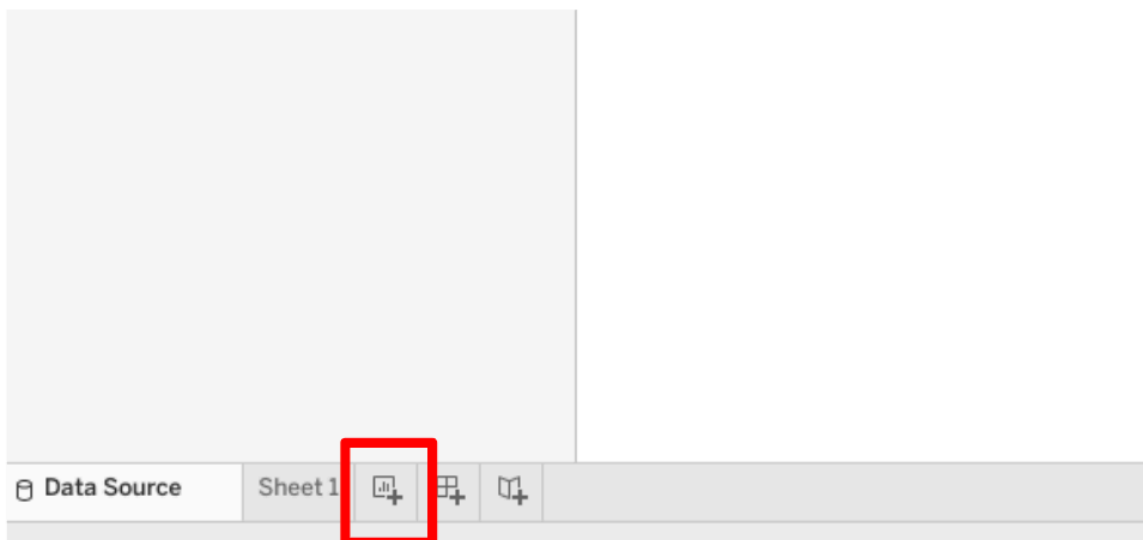
Drag tables here to relate them. [Learn more](#)

tableau_dataset.csv			
4 fields 10479003 rows			
Name			
tableau_dataset.csv			
Fields			
Type	Field Name	Physical Table	Rem...
Date	Date	tableau_dataset...	date
Longitude	Longitude	tableau_dataset...	longit...
Latitude	Latitude	tableau_dataset...	latitude
Number Of Strikes	Number Of Strikes	tableau_dataset...	numb...

tableau_dataset.csv	tableau_dataset.csv	tableau_dataset.csv	tableau_dataset.csv
Date	Longitude	Latitude	Number Of Strikes
6/8/2018	-95.6000	20.0000	1
6/8/2018	-95.2000	20.0000	1
6/8/2018	-83.4000	20.0000	1
6/8/2018	-78.7000	23.0000	1
6/8/2018	-96.3000	21.0000	1
6/8/2018	-85.9000	22.0000	1
6/8/2018	-85.3000	23.0000	1
6/8/2018	-82.3000	23.0000	1

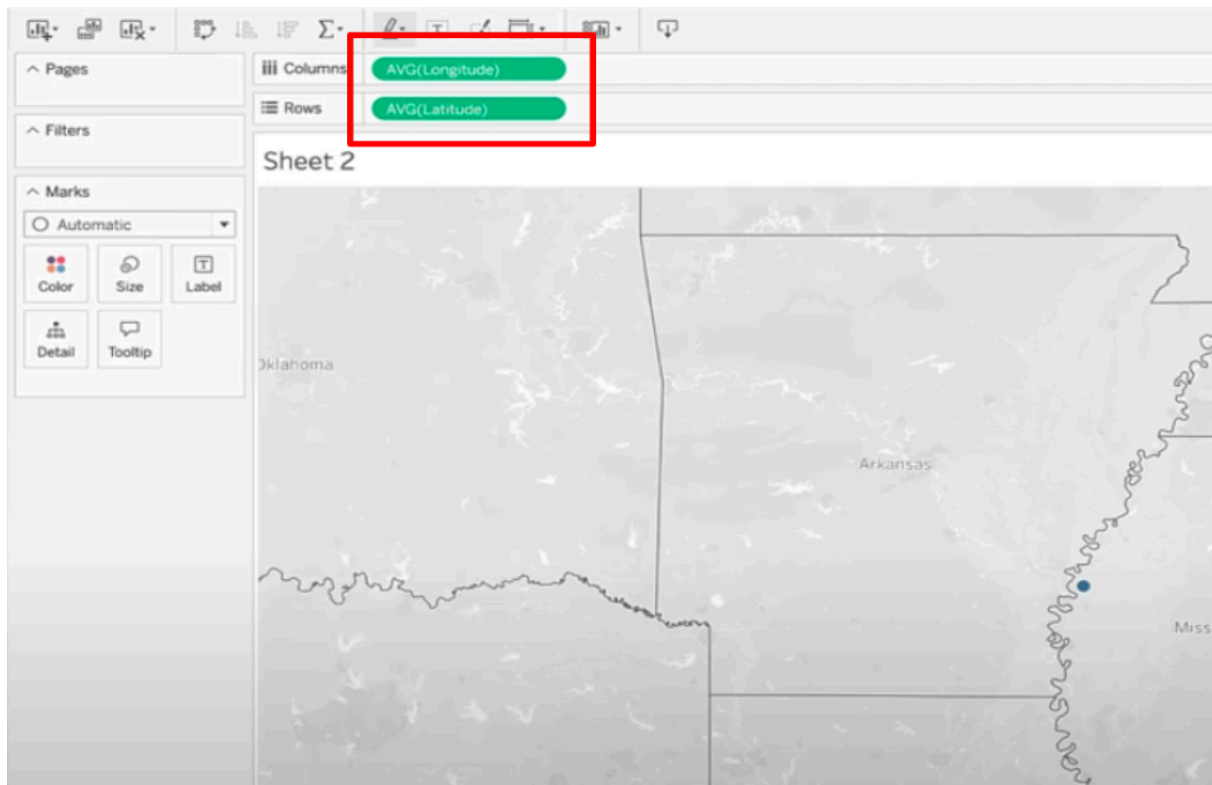
- Click on NEW WORKSHEET.

(Note: Please allow several minutes for data import into a new worksheet)

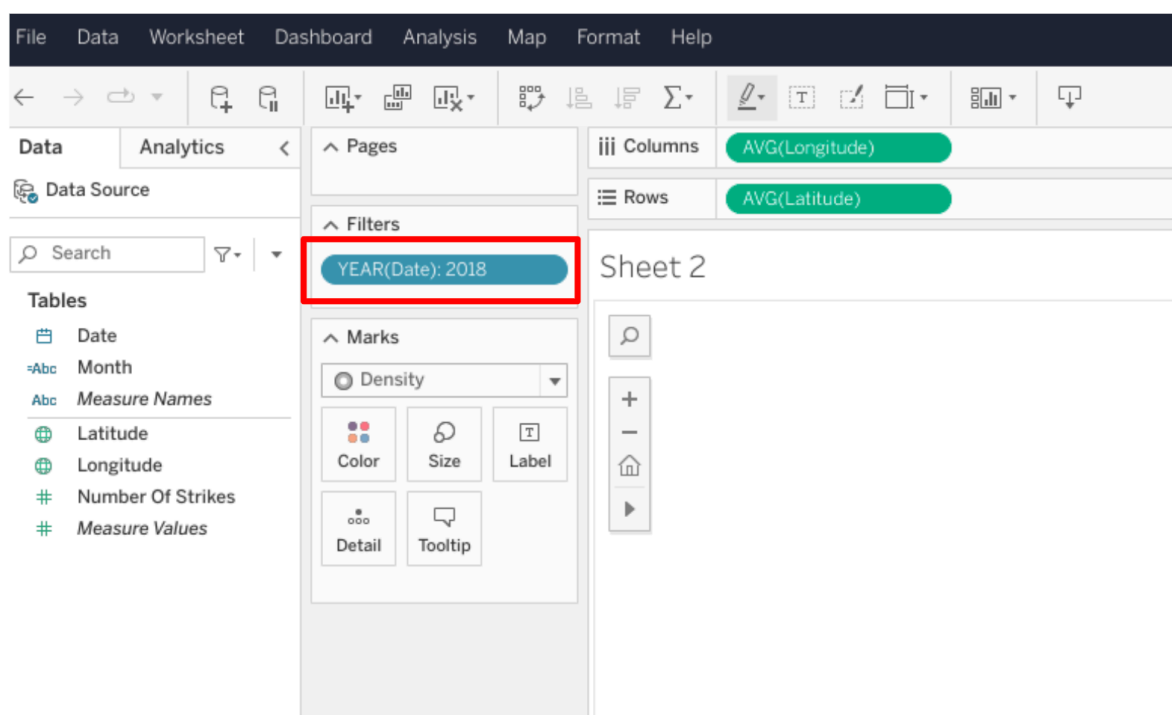


- Drag LONGITUDE into the column field. Drag LATITUDE into the rows field.

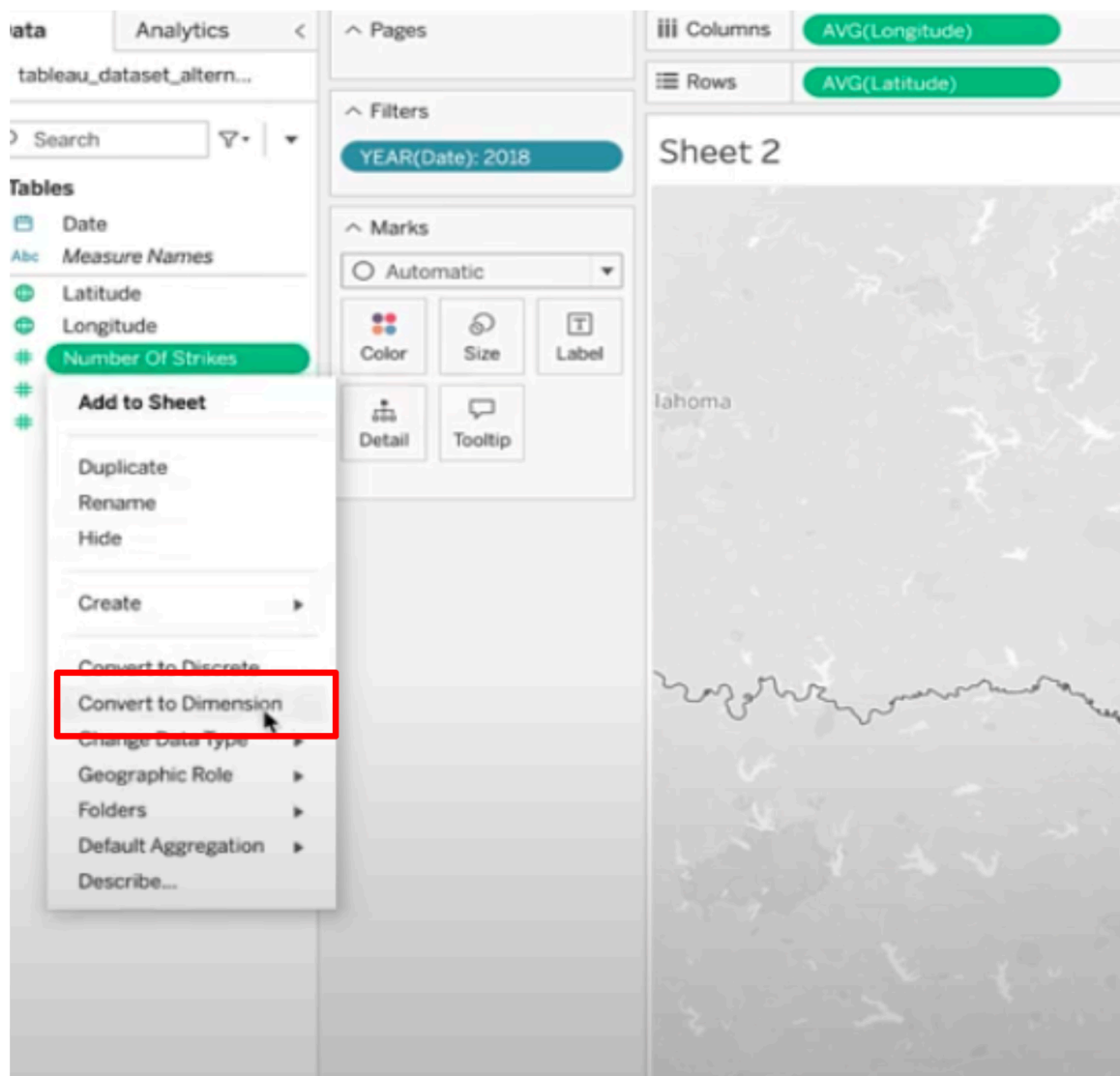
(Note: Make sure the latitude and longitude fields are set to continuous dimensions)



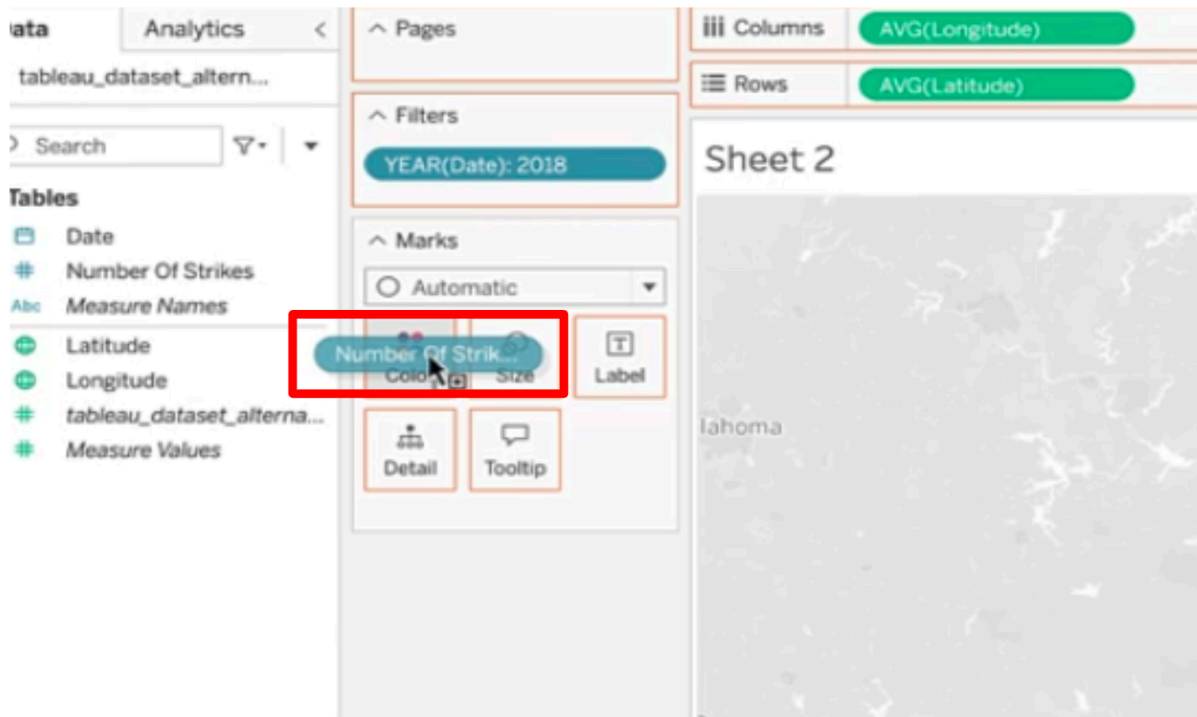
- Drag DATE into Filters field. Filter to only 2018.



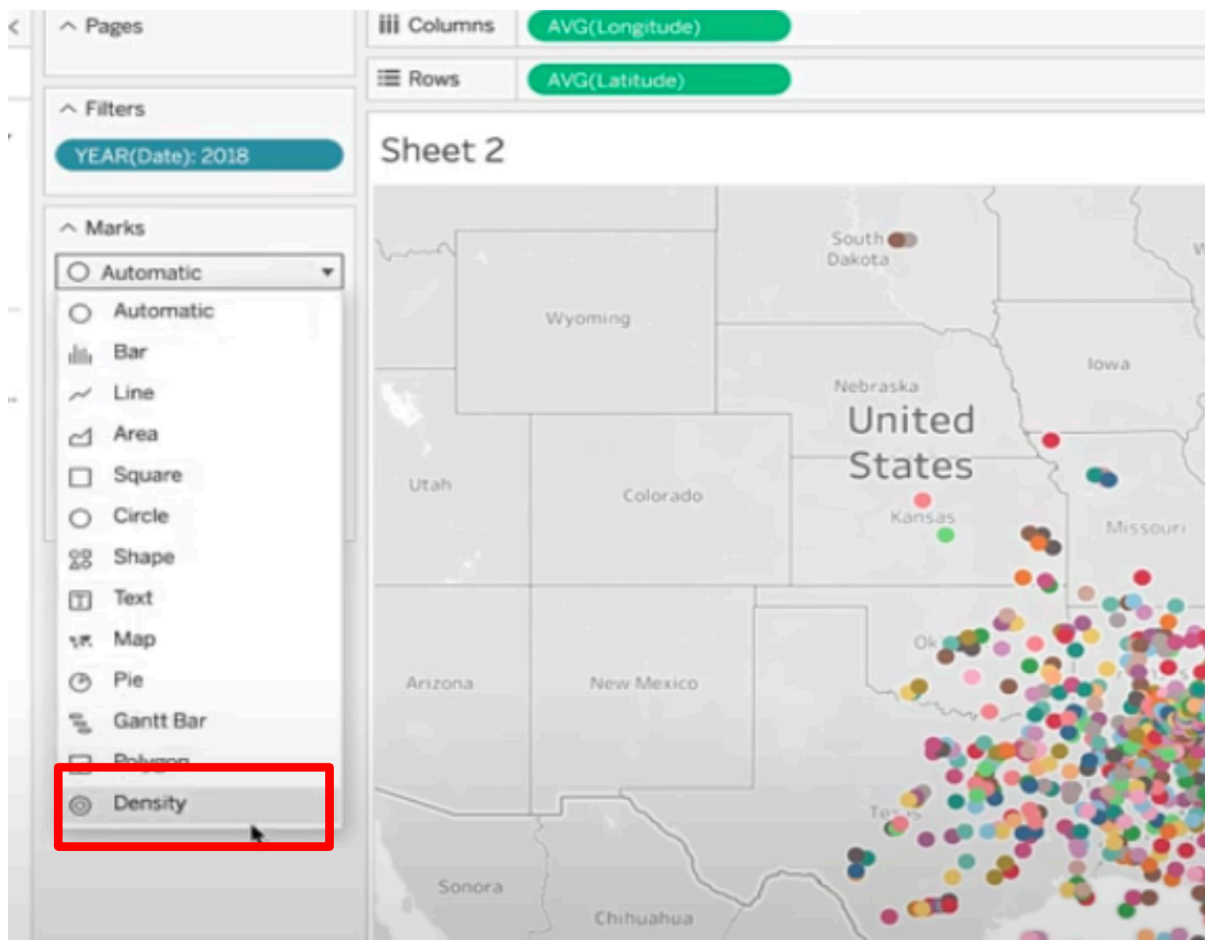
- Click on NUMBER OF STRIKES dropdown.
- Select "Convert to Dimension."



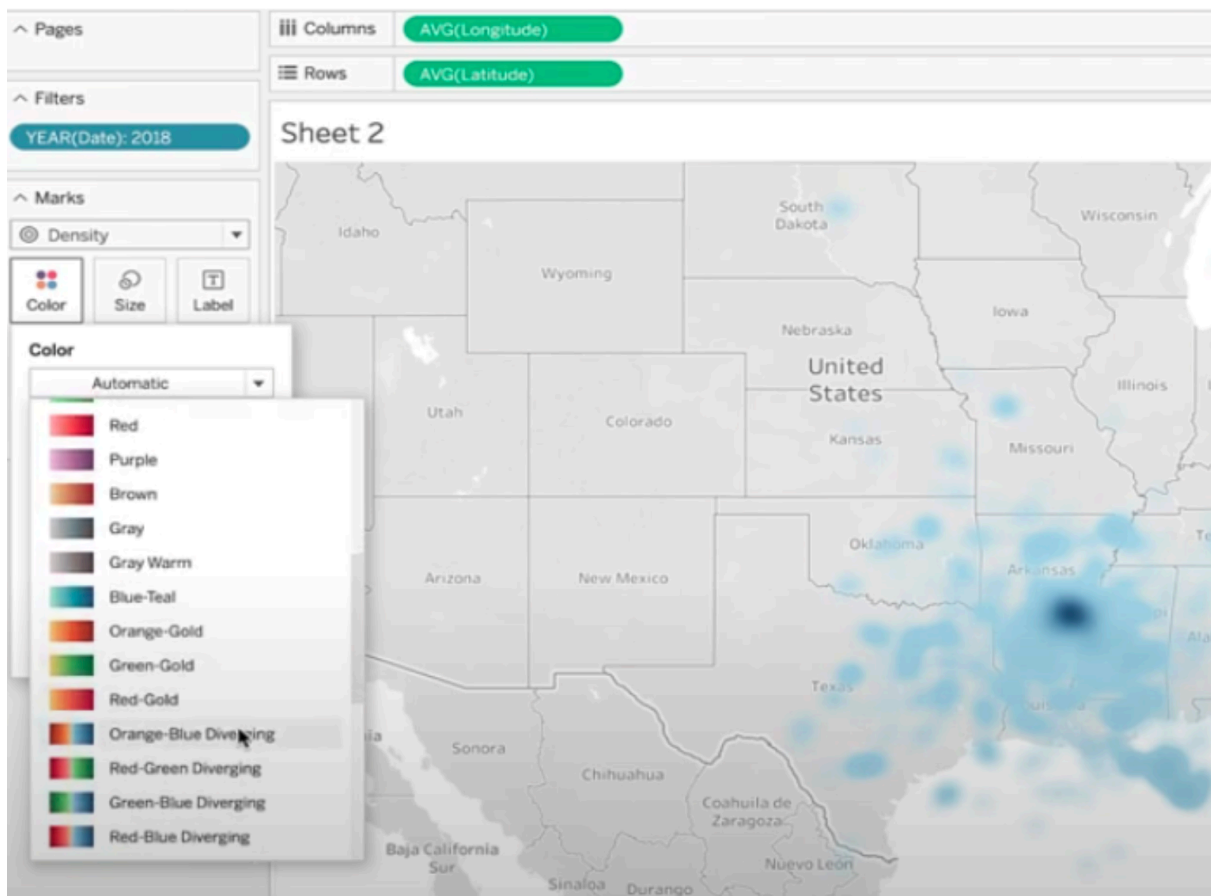
- Drag NUMBER of STRIKES to the box labeled COLOR.



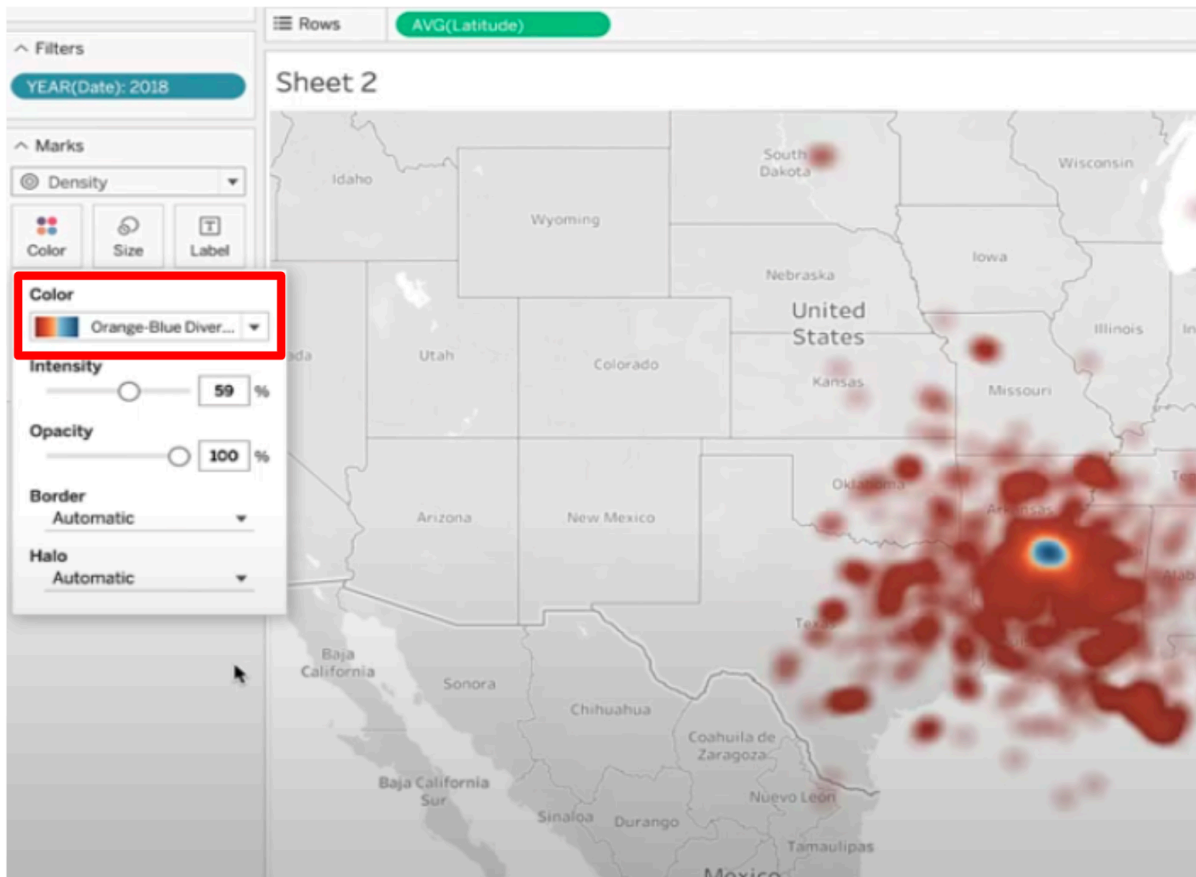
- Select Density in MARKS dropdown.



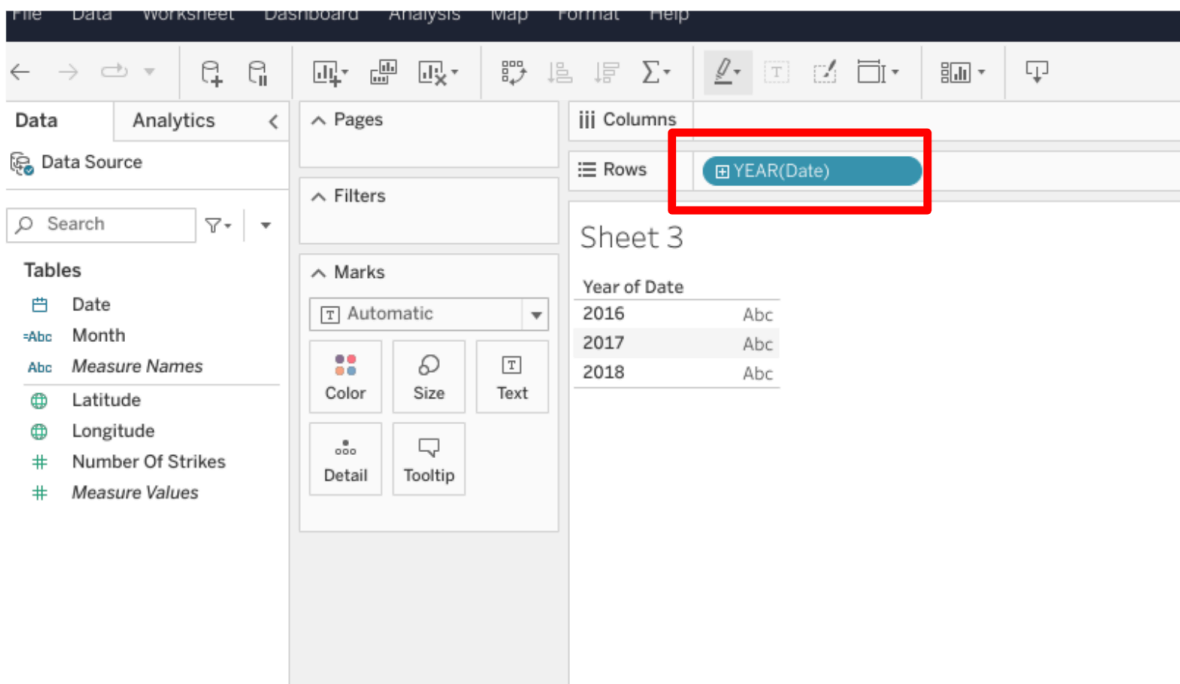
- Click on color and show the dropdown. Select any color you'd like.



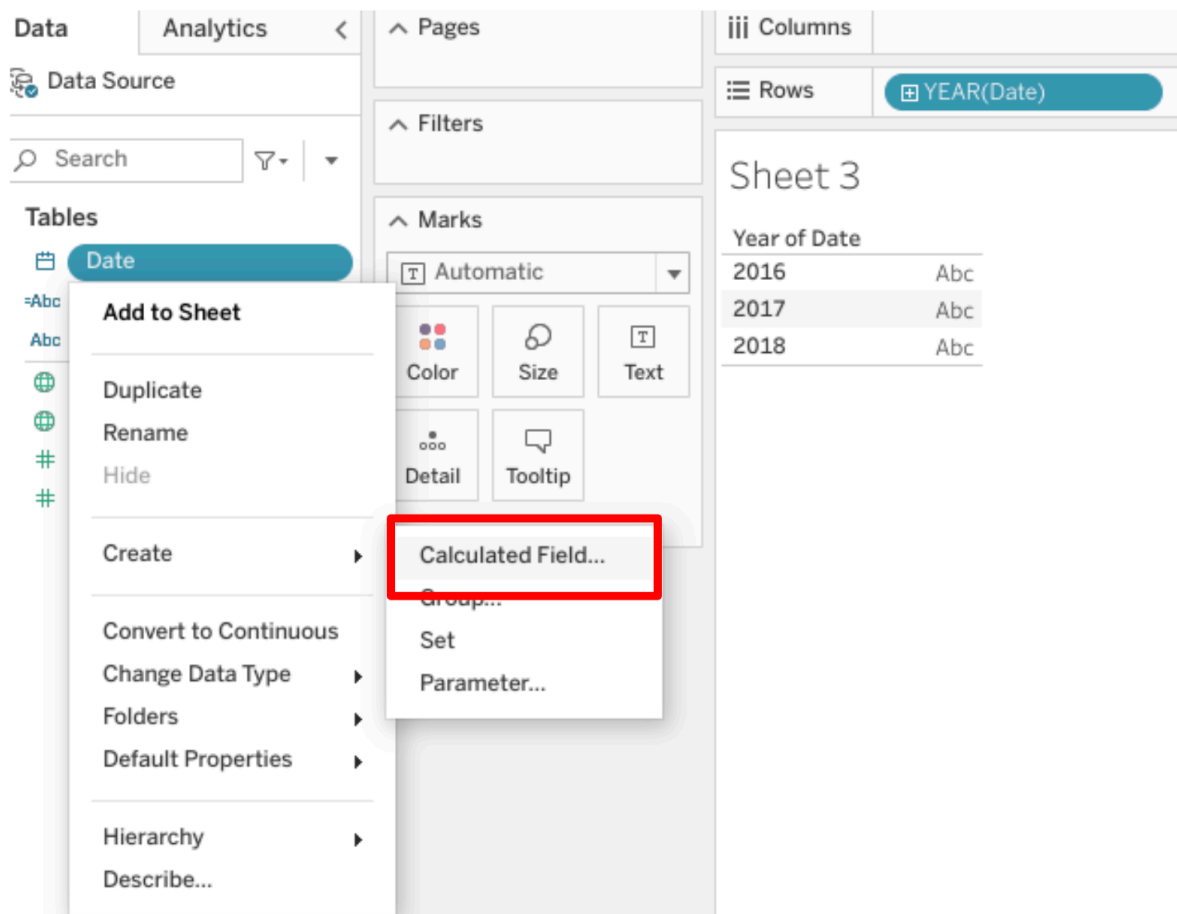
- And here is your geographic map of the location of lightning strikes in the U.S. in 2018.



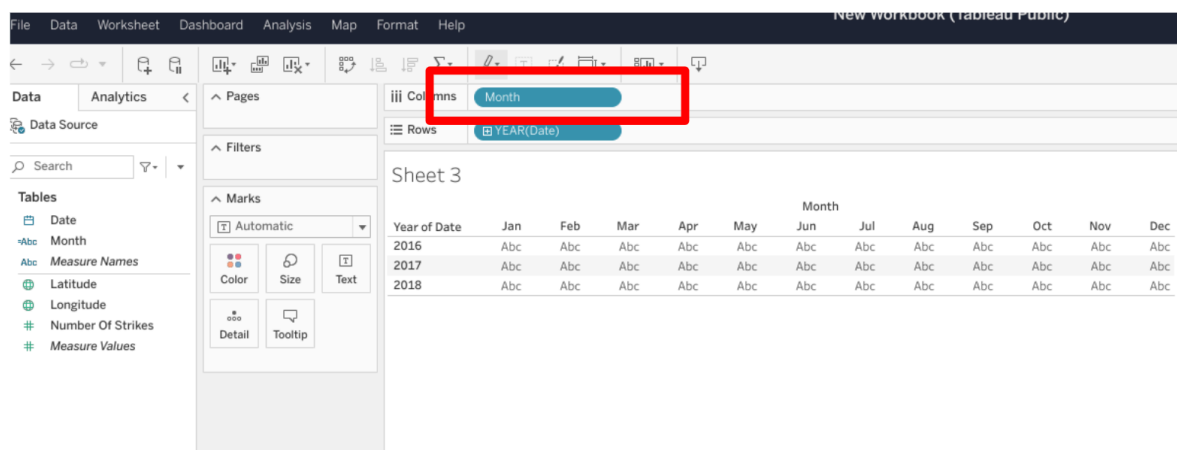
- Click on "new worksheet."
- Drag DATE to the ROW field and select YEAR.



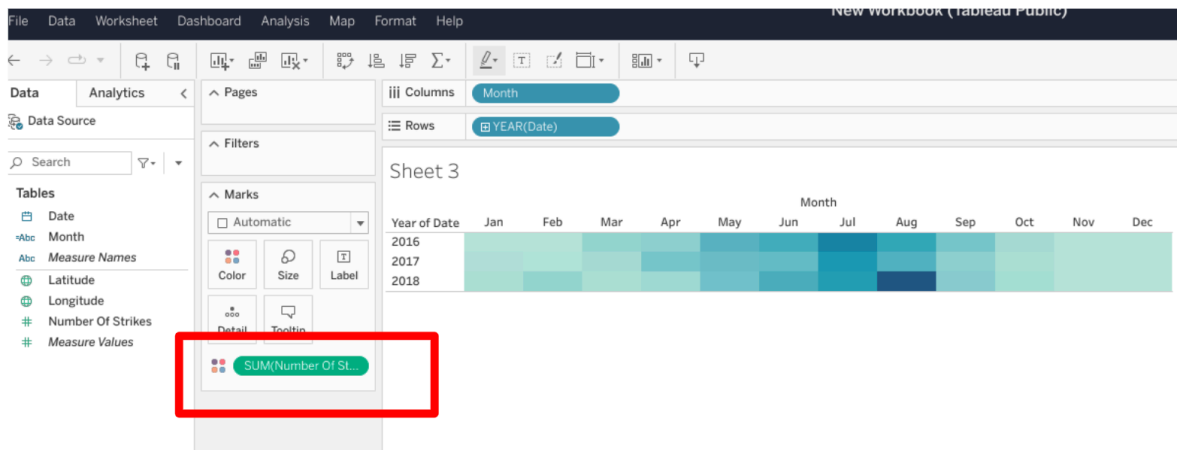
- Click on DATE dropdown in the "Tables" menu. Click CREATE and select Calculated Field.



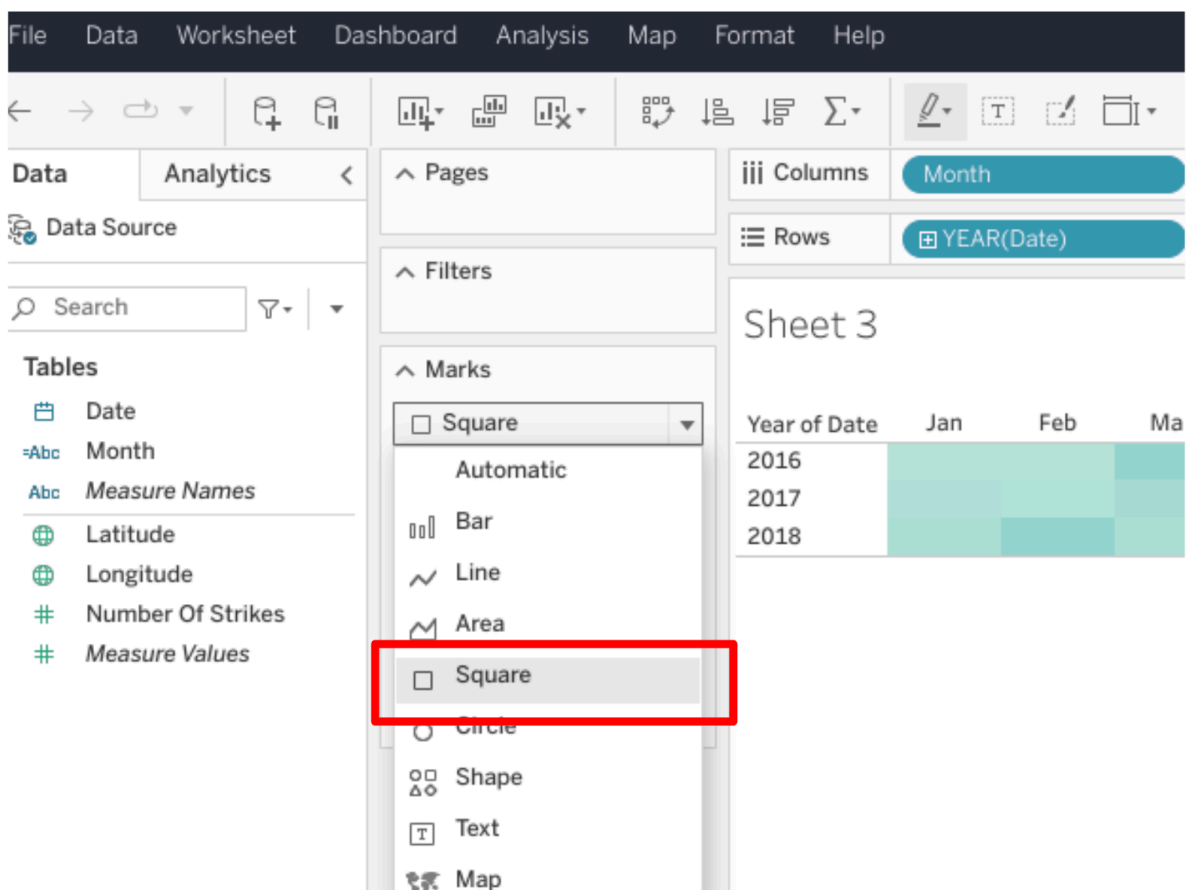
- Type in Month for Calculated Field name.
- Type into field:
 - **LEFT(DATENAME('month',[date]),3)**
 - Click Apply.
 - Click OK.
- Drag month to column field.



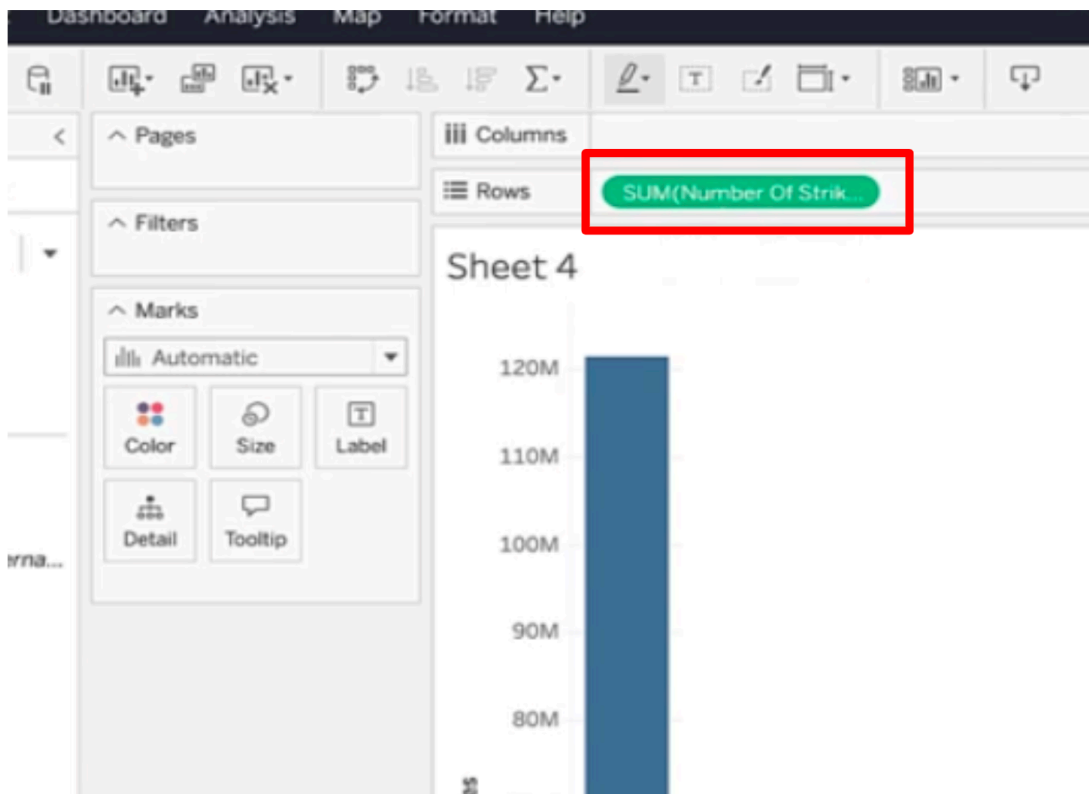
- Drag NUMBER of STRIKES to the color square under the MARKS field.
- Make sure NUMBER of STRIKES is converted to Measure. To do so, right click on Number of Strikes field and select 'Convert to Measure.'



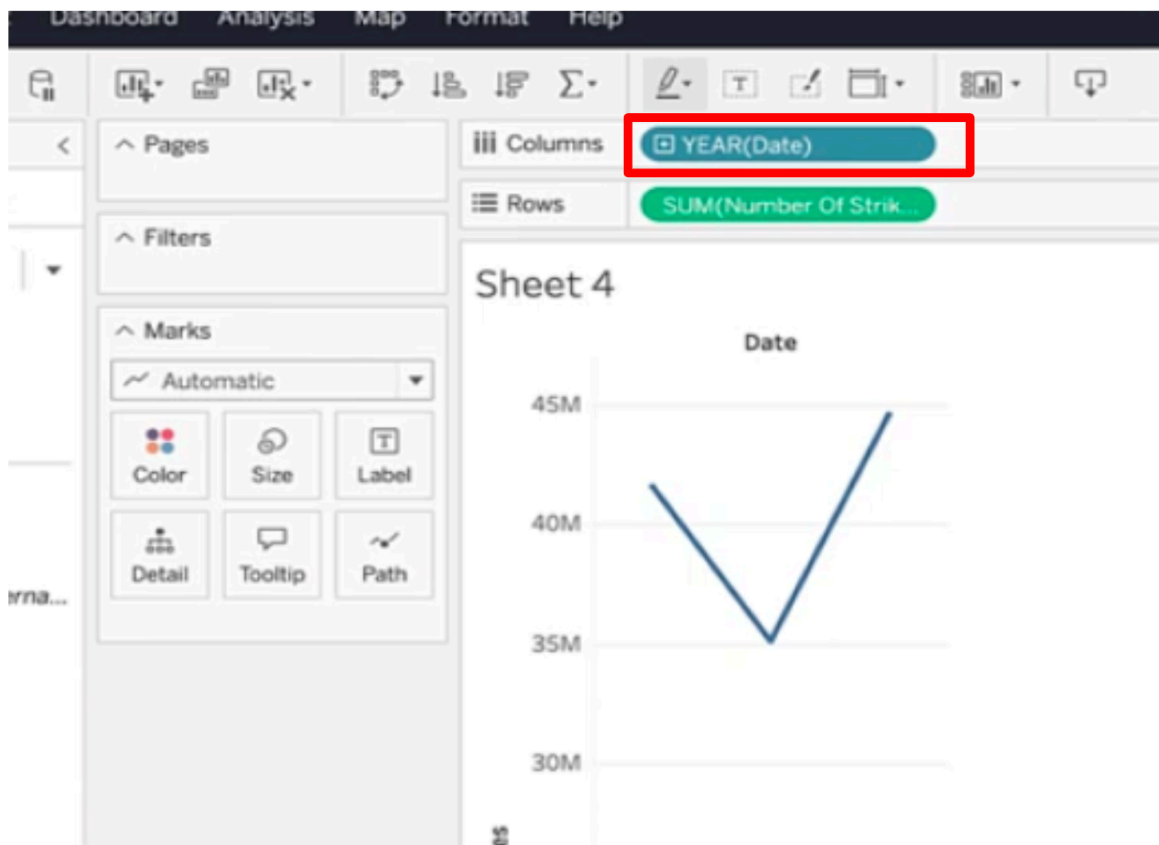
- In Marks dropdown, select SQUARE.



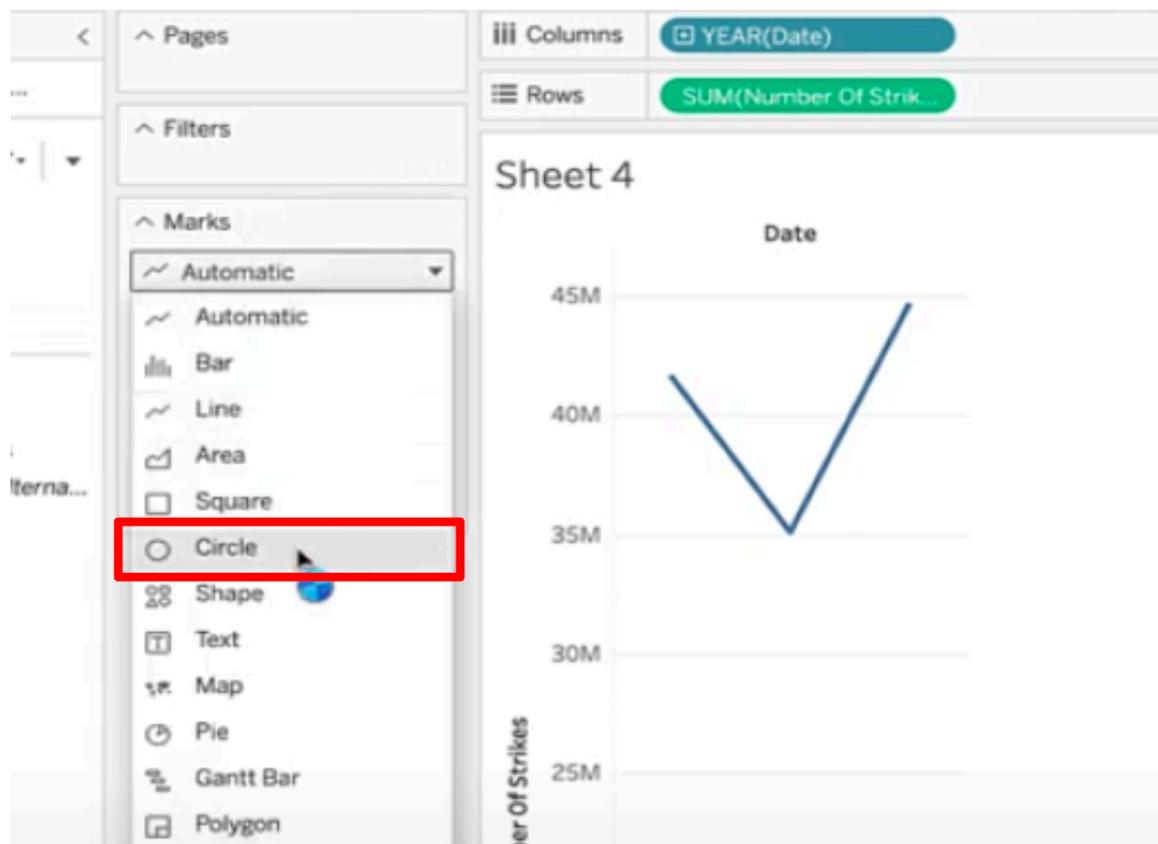
- Feel free to change the color. You've completed a heatmap.
- Click on NEW WORKSHEET.
- Drag NUMBER OF STRIKES to the rows field.



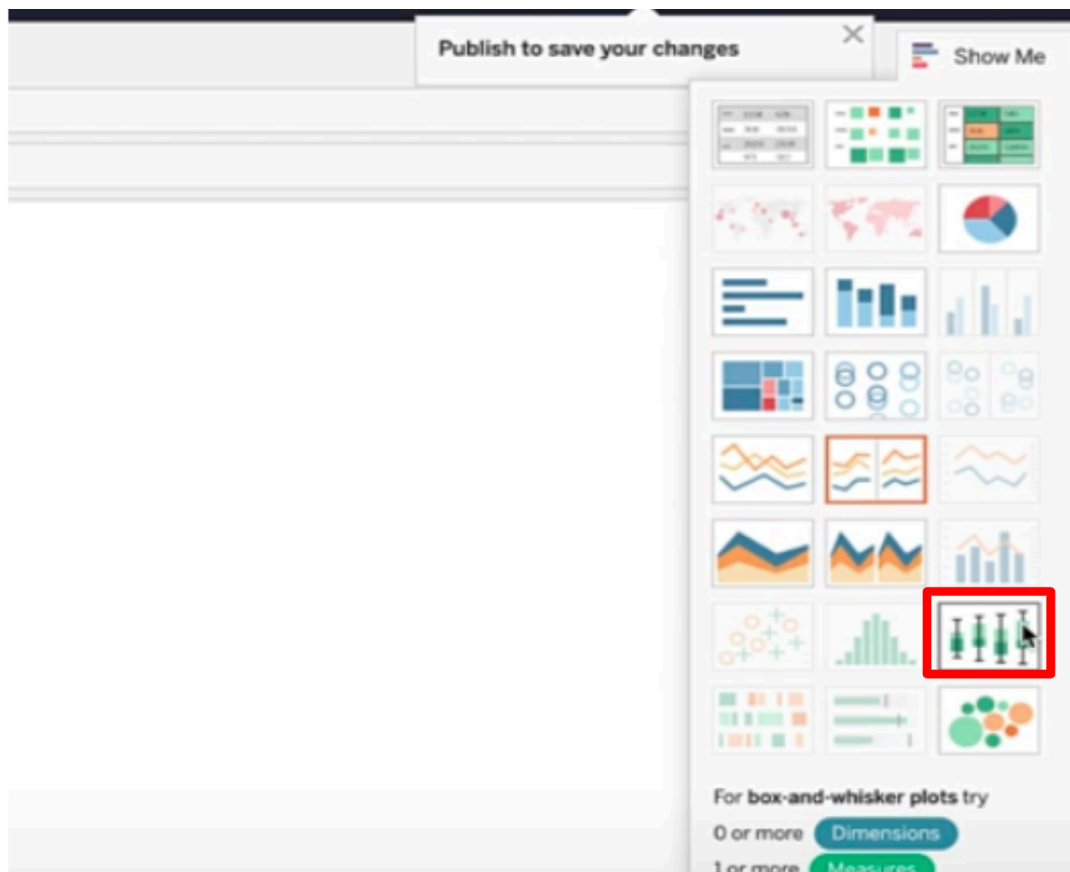
- Drag DATE to the column field and select YEAR.



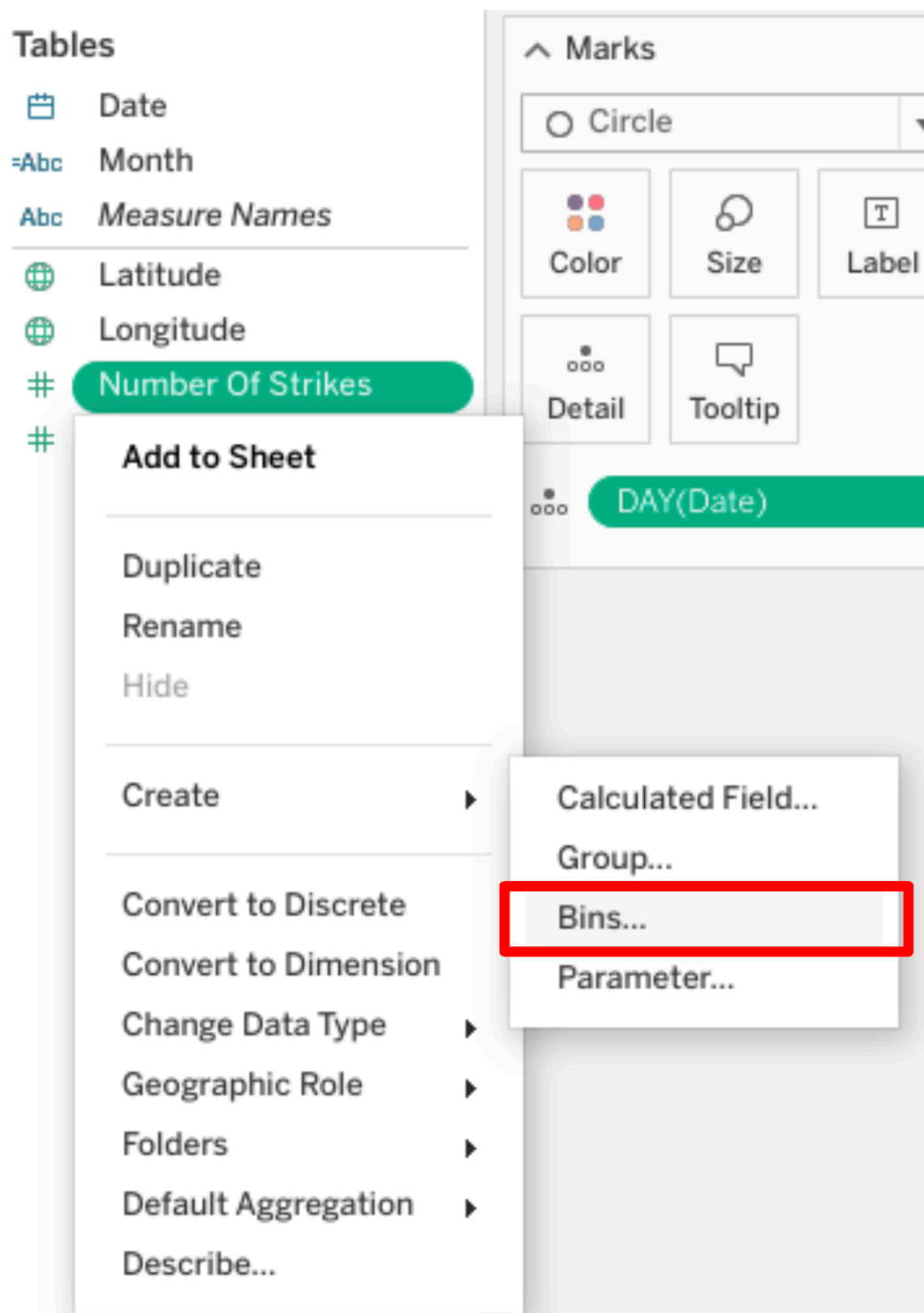
- **Note:** If nothing is shown in the Columns field, drag YEAR back to the Columns filter again.
- Select CIRCLE from MARKS dropdown.



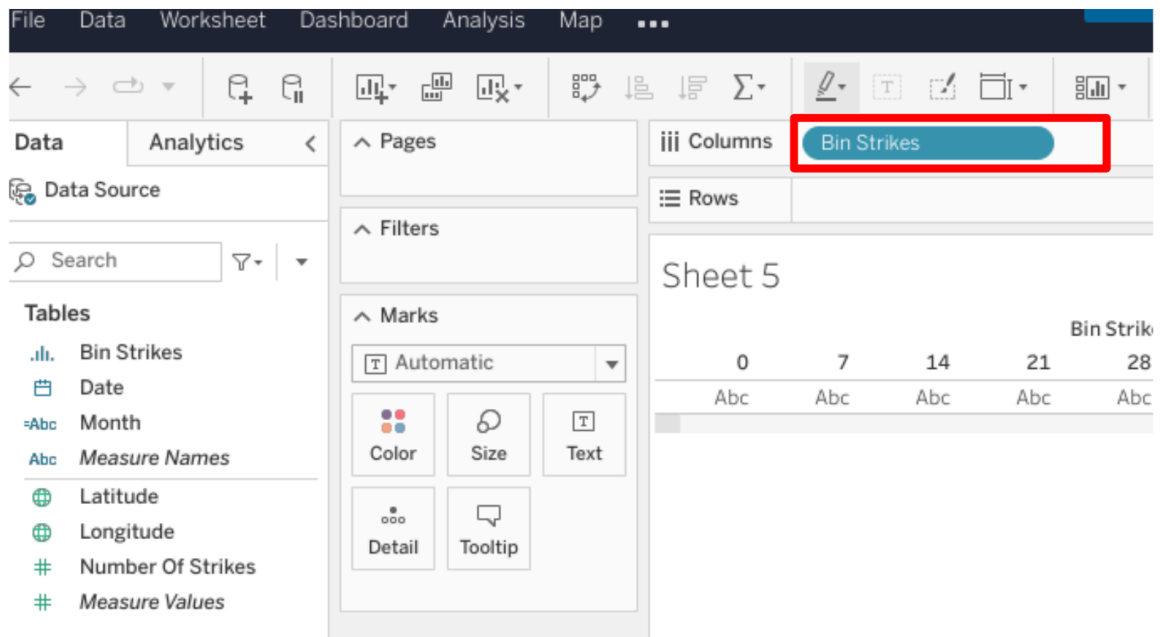
- **Note:** If box plot isn't the default, select the BOX and WHISKER PLOT from the SHOW ME dropdown.
- Drag Date to the column field again, if that has been removed.



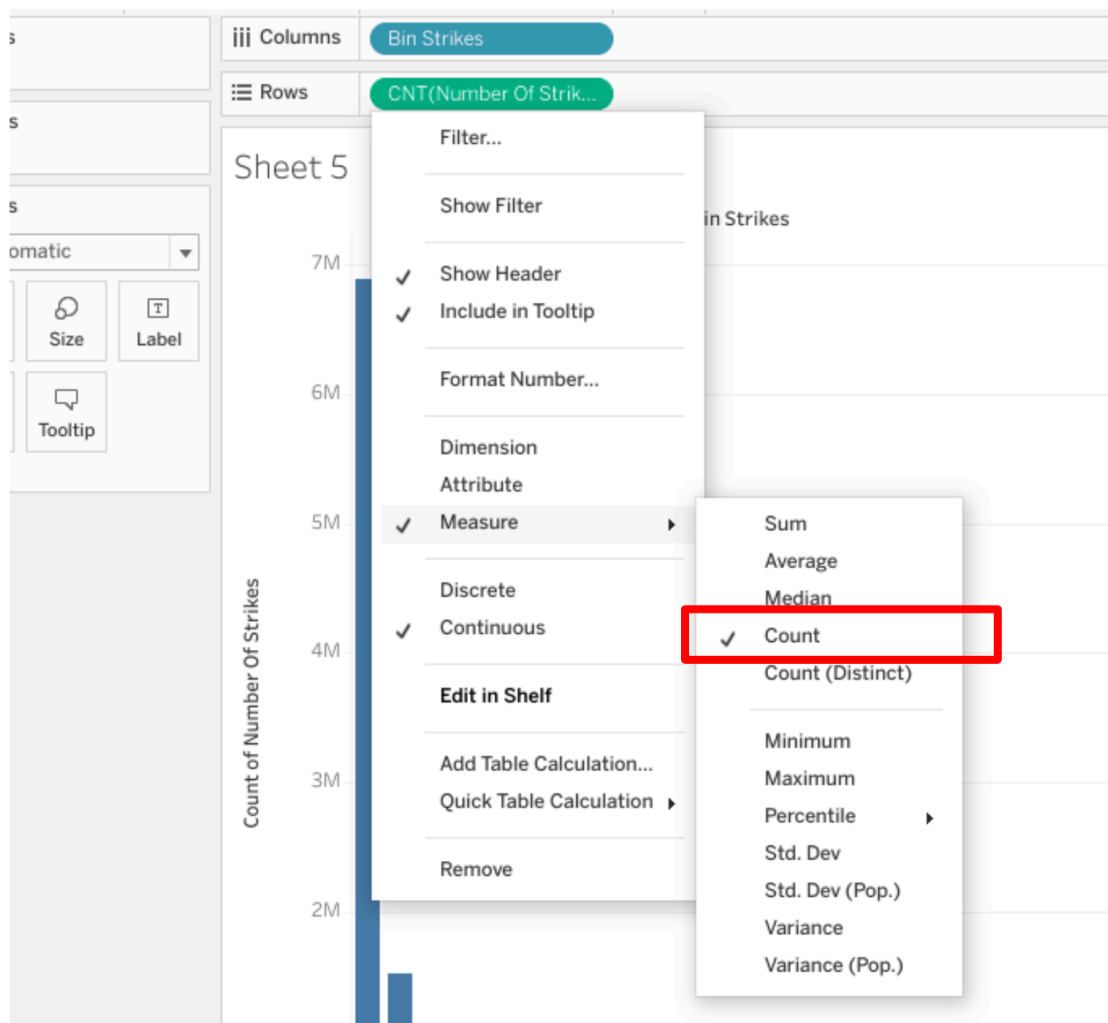
- Drag DATE into the detail square under the MARKS field, and select DAY. **Note:** When you do this, you'll see that Tableau changes your YEAR(Date) filter to QUARTER(Date) to show an additional level of detail. That change is shown in the instructional video, so please proceed.
- Now you've completed a boxplot.
- Click on NEW WORKSHEET.
- Click on the dropdown of NUMBER of STRIKES in Table menu, and select CREATE and then BINS.



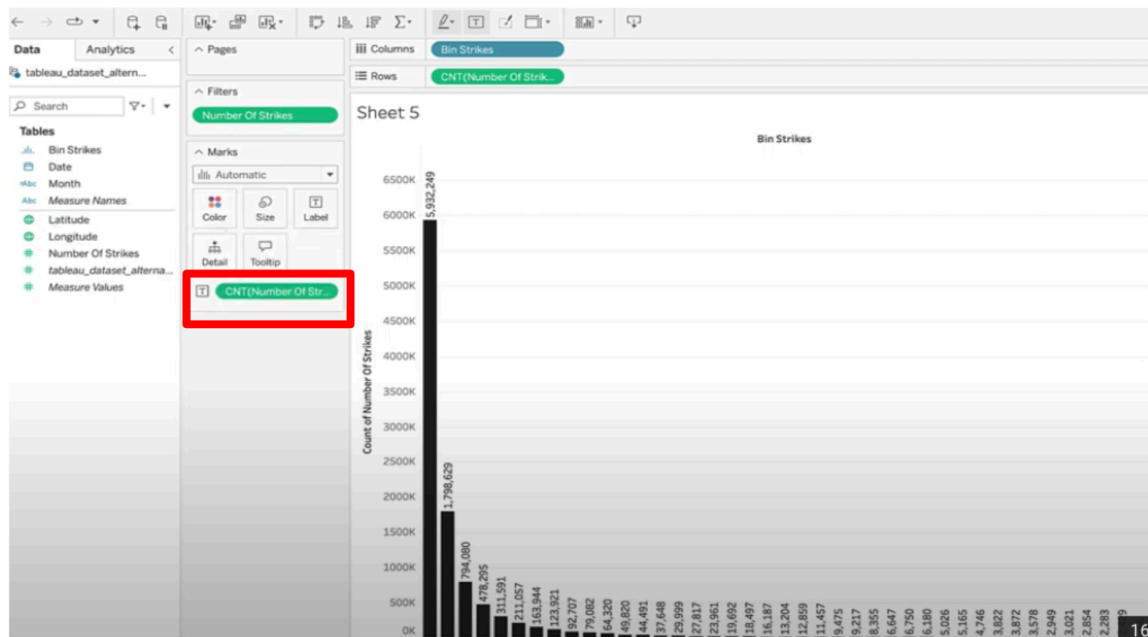
- Give bin a name: BIN STRIKES.
- Select a number between 5 and 10 for "Size of Bins" field.
- Drag BIN STRIKES to columns field.



- Drag NUMBER OF STRIKES to Row Field.
- In NUMBER of STRIKES dropdown, make sure COUNT is selected.



- Drag NUMBER of STRIKES to filter field.
- Limit the field numbers between 1 and 200.
- Drag NUMBER of STRIKES to the LABEL.
- Select COUNT.
- Change color as desired.
- Now you know how to create a histogram!



Activity Exemplar: Design a bar graph that tells a story in Tableau Public

Here is a completed exemplar along with an explanation of how the exemplar fulfills the expectations for the activity.

Completed Exemplar

To review the exemplar for this course item, click the link below.

Link to exemplar: [Seasonal Seoul Average Bike Rentals on Weekdays in 2018](#)

Assessment of Exemplar

Compare the exemplar to your completed activity. Review your work using each of the criteria in the exemplar. What did you do well? Where can you improve? Use your answers to these questions to guide you as you continue to progress through the course.

Note: *The exemplar represents one possible way to complete the data visualization. Yours will likely differ in certain ways. What's important is that your data visualization meets the business scenario criteria.*



How the exemplar meets the criteria

Notice how the exemplar posted to Tableau Public uses the detail from the given scenario to adhere to the following guidelines:

The data visualization includes only data from 2018.

- The "YEAR(Date)" variable is in the filter field and is check marked for 2018 only.

The data visualization includes only weekdays, Monday to Friday.

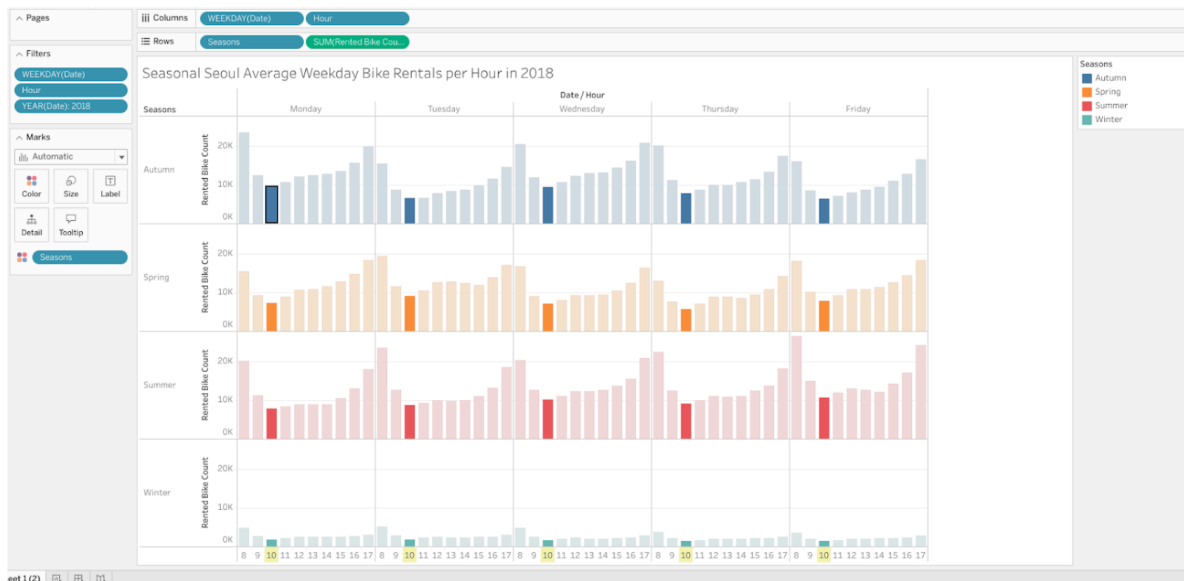
- The "WEEKDAY(Date)" variable is in the filter field and is check marked for Monday through Friday only.
- The "Hour" variable in the filter field is check marked on the hours 8 to 17.

The data visualization type makes sense for the data shown and includes color, contrast, labeling, and emphasis elements that focus on the most relevant data.

- The chart selected was a bar chart. A scatter plot and box plot could be helpful, but would likely be too complex for the purpose of the visualization. A geographic map would not be helpful at all since there is no geographic data.
- The blue color bars represent viable times of day and dark orange are times in which bike rental traffic is too high for maintenance to be viable.
- Annotations are added to highlight the relevant information.
- A title was added.
- The "Seasons" variable was added to provide a more thorough answer.

The data visualization can be observed without scrolling out or zooming in.

- Because of the filtering mentioned above (2018 data, excluding weekends and non-business hours), the visualization fits on one screen or a printout.



Follow-along guide: Craft compelling stories with Tableau

This document includes detailed instructions for how to perform the data visualizations described in the video "Craft compelling stories with Tableau."

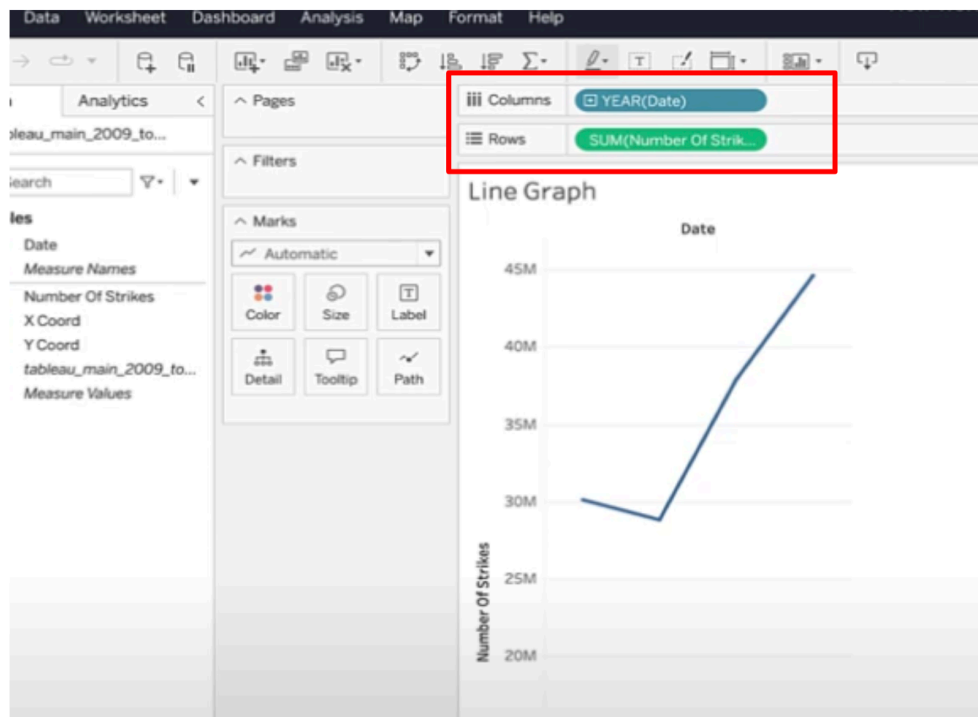
The following guide points out areas of the video that may require adjustment. These resource guides can also serve as a set of usability reminders for you to recall when using Tableau in your future career.

Instructions

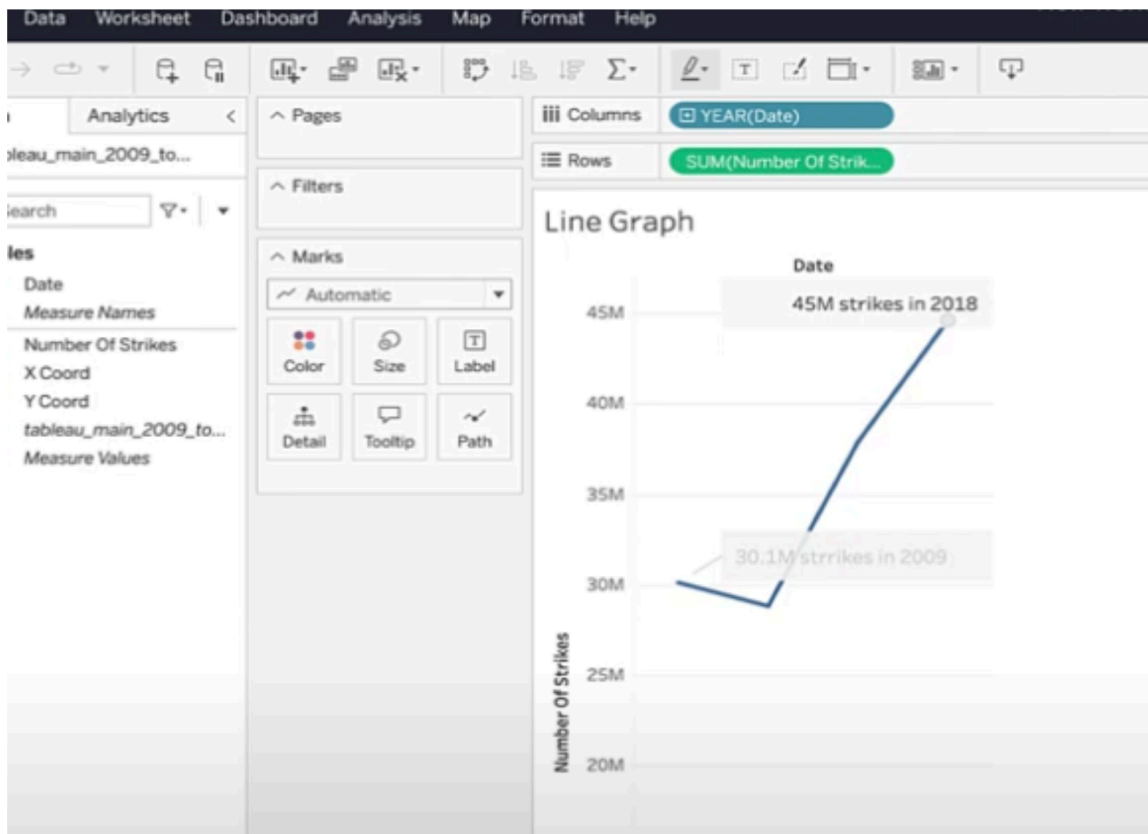
- Go to <https://public.tableau.com/s/>
- Since you've already set up your Tableau Public profile, all you need to do is log in and select **Web Authoring** under **Create** in the navigation bar.
- Select the appropriate CSV file provided in the instructions. The dataset you'll use with this instructional video is: `tableau_main_2009_to_2018.csv`.
 - Please be aware that when you download the zip file folder provided, the computer automatically names that zip file folder with a long string of numbers and letters. You have to open that folder and then upload the individual files that are named correctly and match what's shown in the video.
- Click on NEW WORKSHEET.

Note: Please allow several minutes for data import into a new worksheet.

- Move the blue Date field to the column field.
- Select YEAR from dropdown.
- Move NUMBER OF STRIKES to the row field.

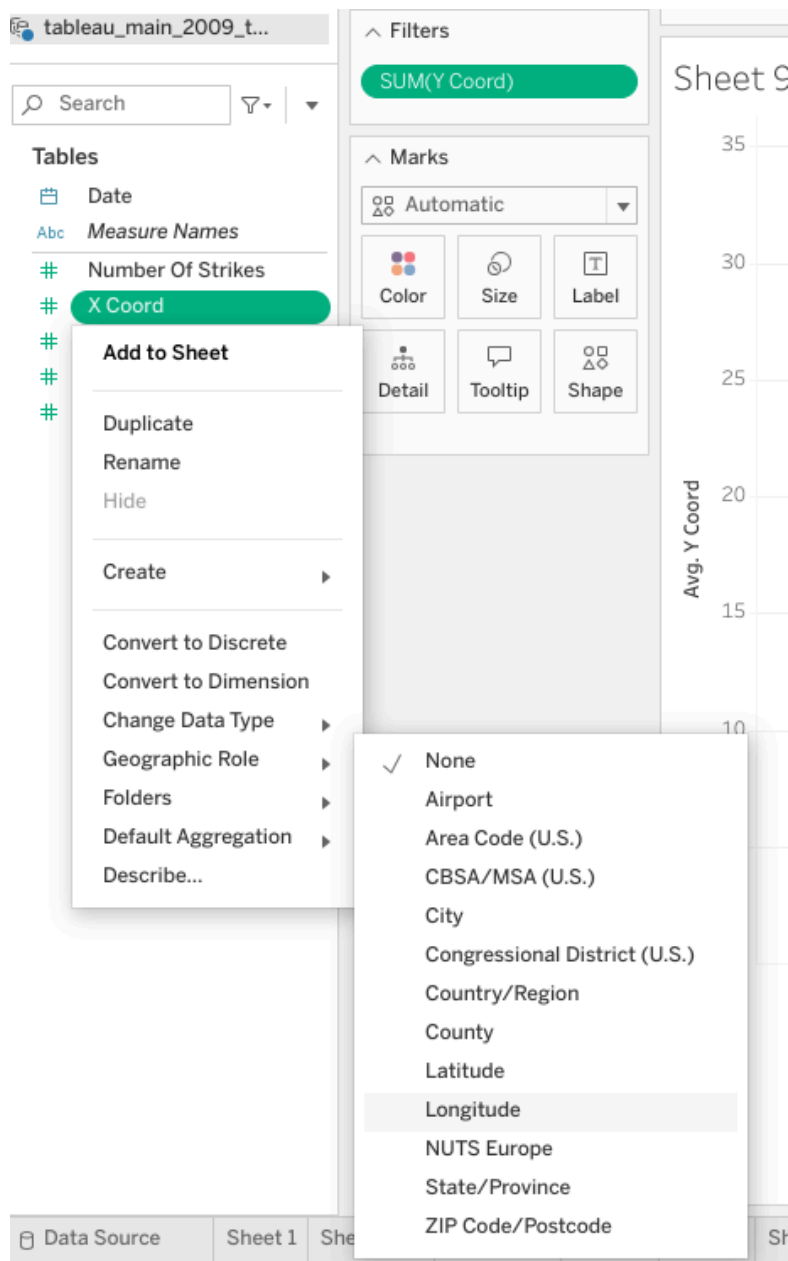


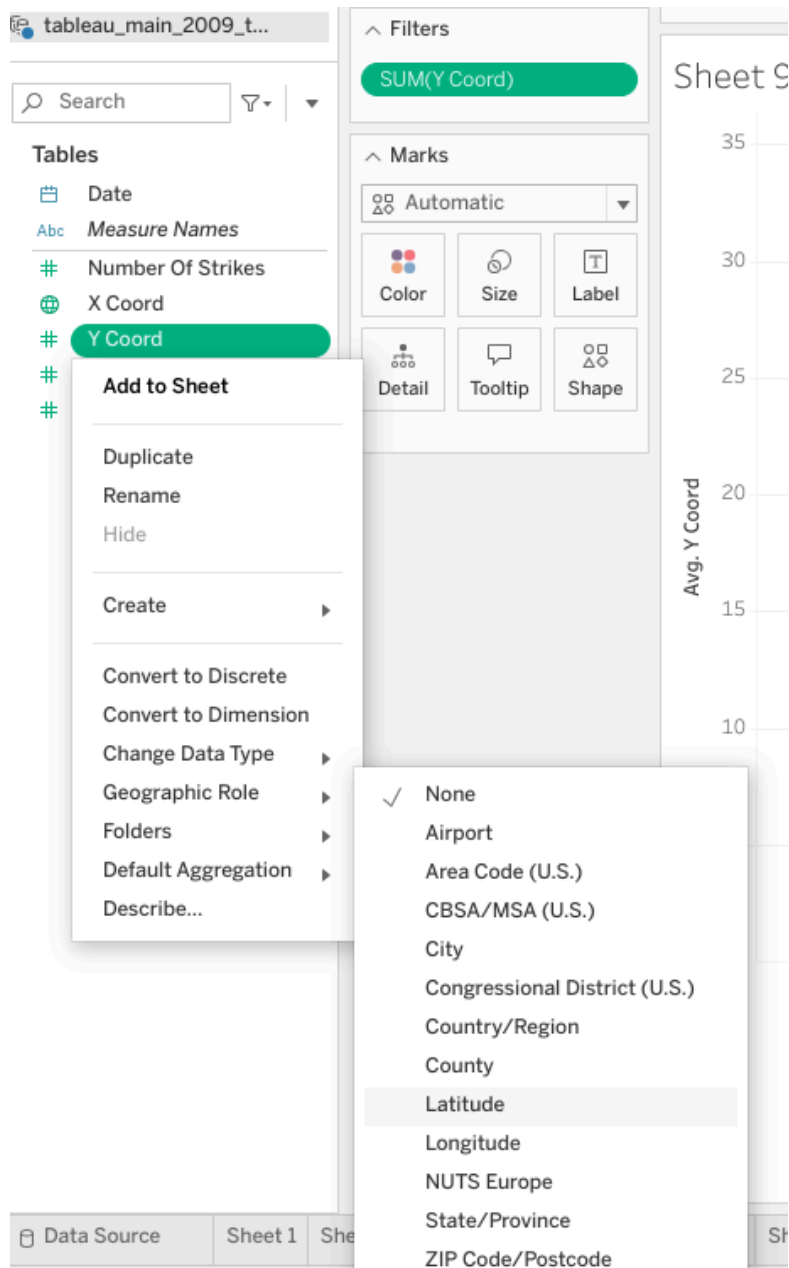
- If your visualization doesn't default to a line graph, click on "SHOW ME" in the upper right of the screen and select Lines (discrete).
- Create annotation by **right clicking** on each end of the line and selecting "Annotate mark."
- Type in "30.1M" for the Number of Strikes.
- Create annotation on the final mark in the same way.
- Type "45M" for number of strikes in 2018.



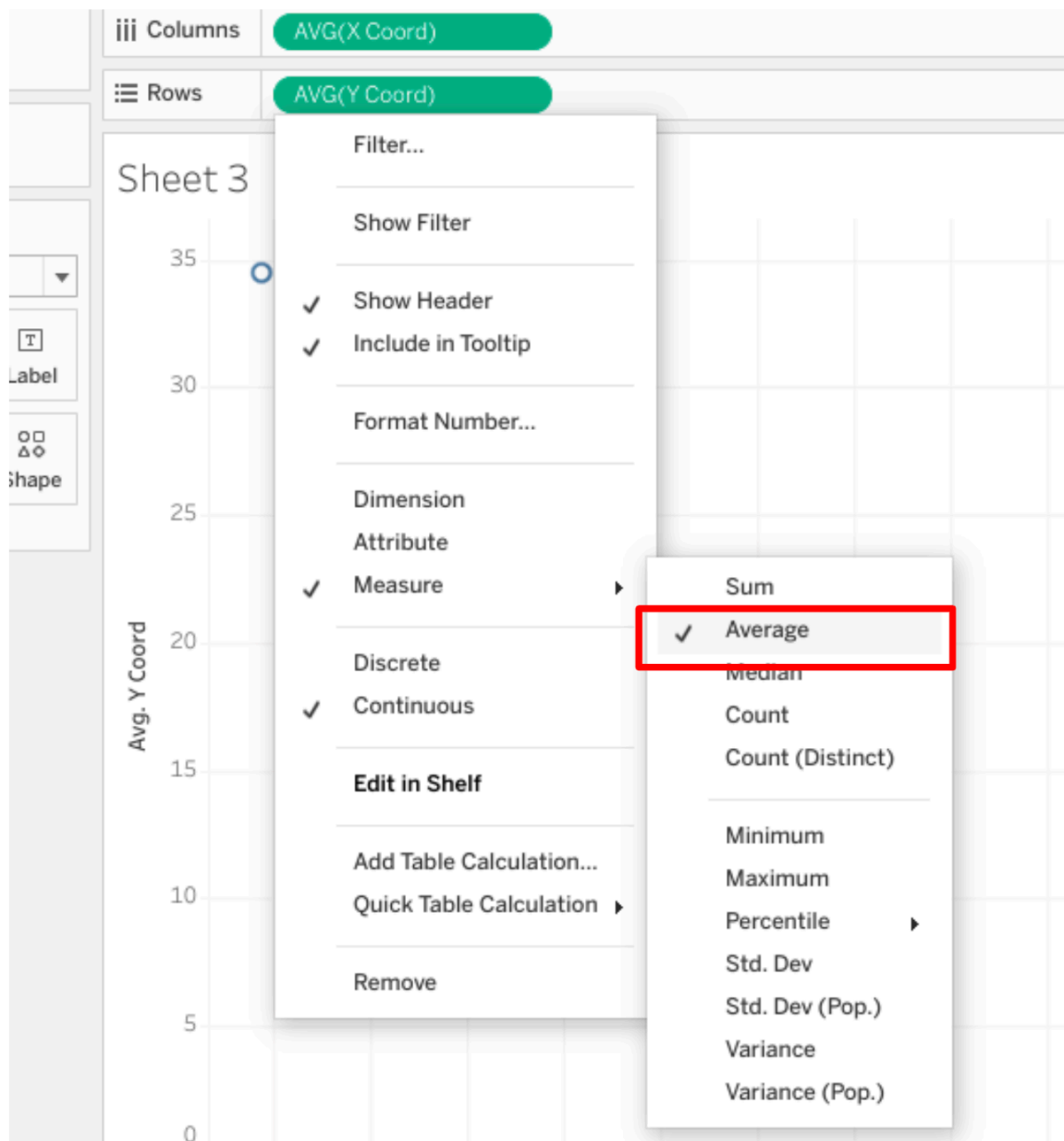
- Click on NEW WORKSHEET.
- Drag the X-coord to the column field. Drag the Y-coord to the Row field.

Note: Make sure the X and Y coordinates are continuous dimensions with Y coord geographic role set to latitude and X coord geographic role set to longitude.

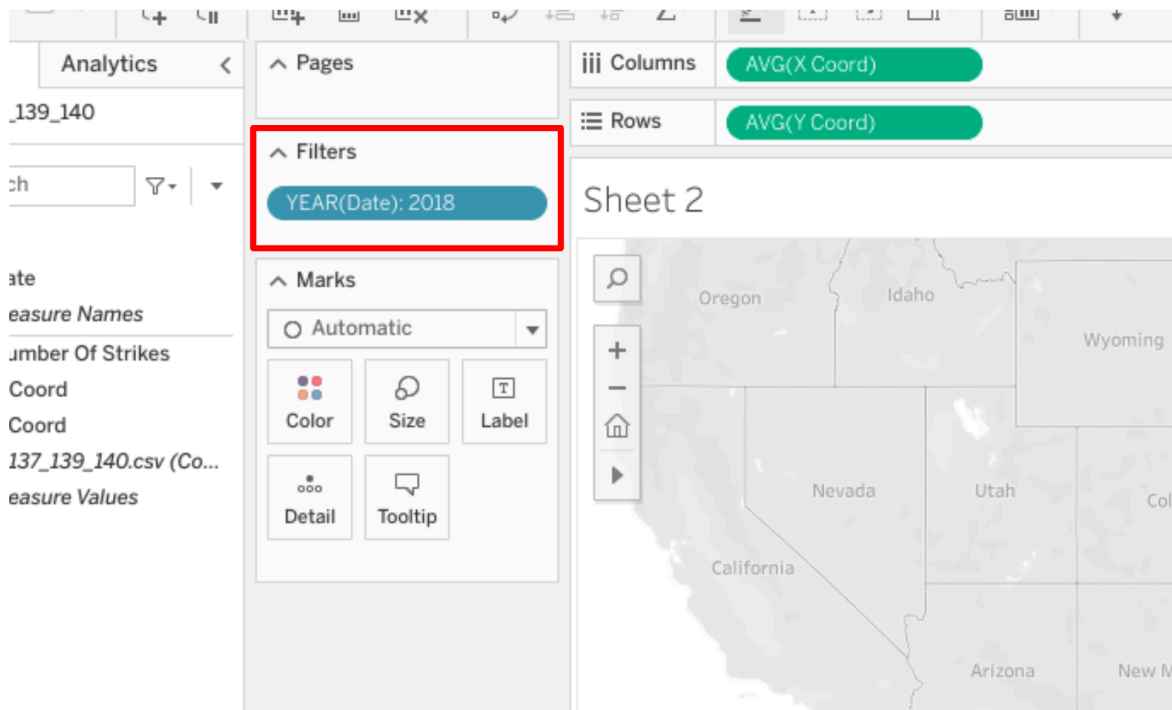




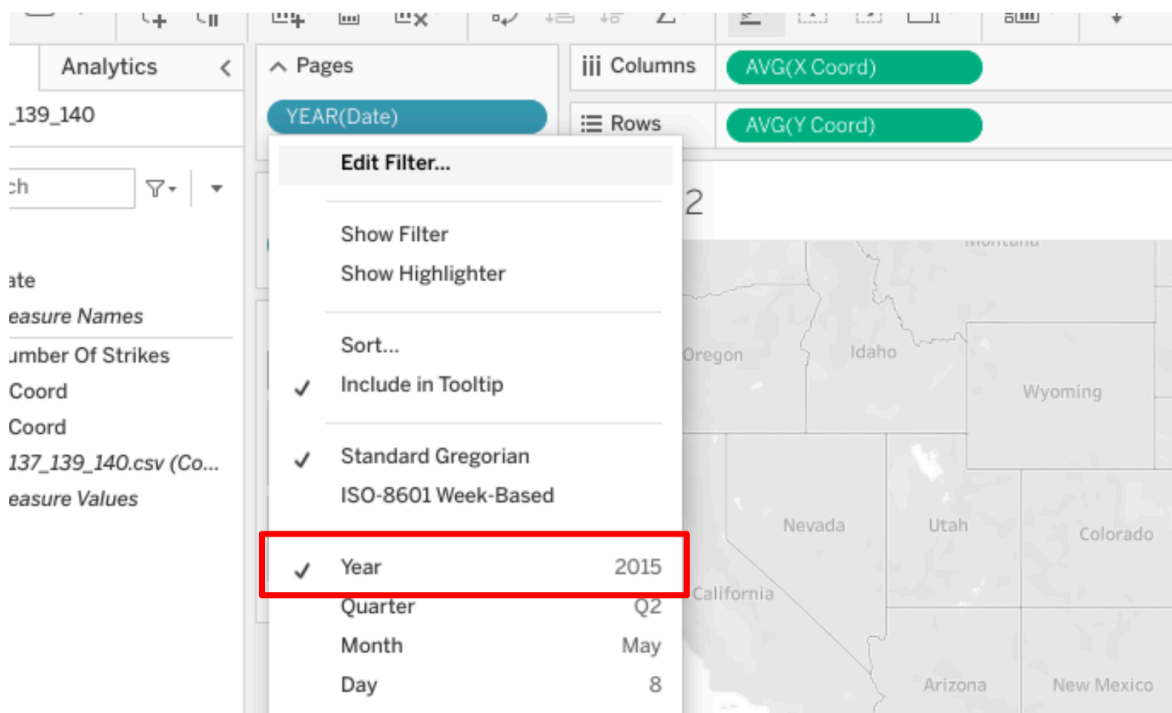
- Click on the dropdown for both the X and Y coordinates.
- Select MEASURE then AVERAGE.



- Drag DATE to filter field.

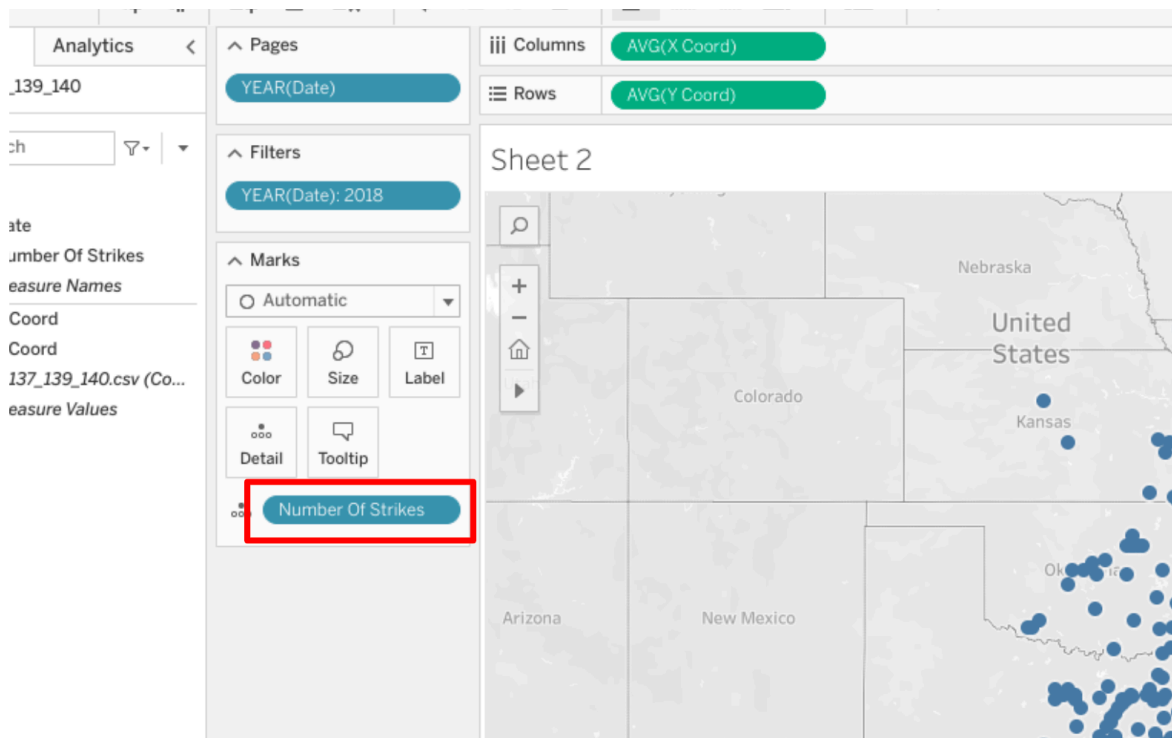


- Drag DATE to pages field, select year from dropdown.

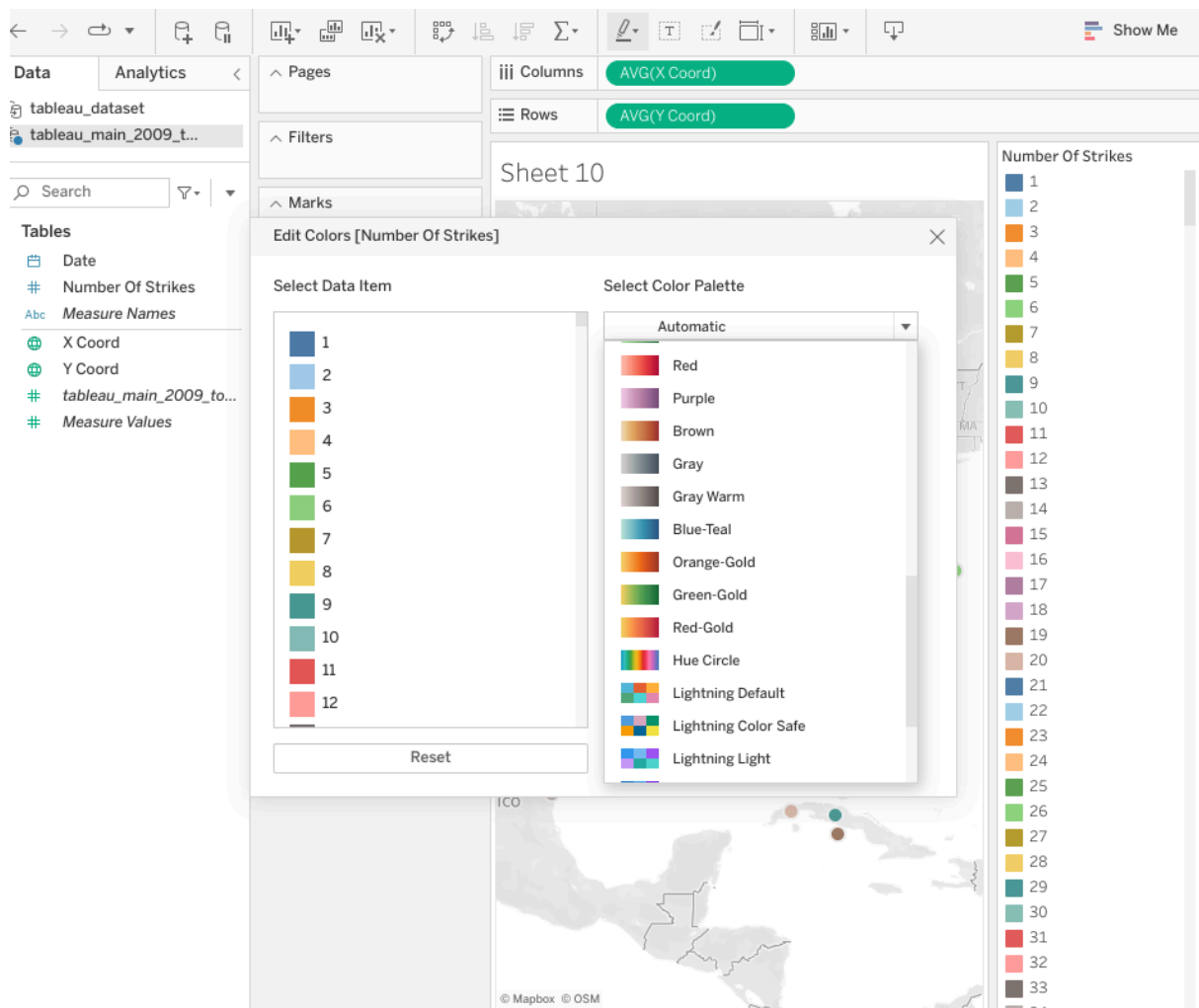


- Drag NUMBER of STRIKES to the detail field.

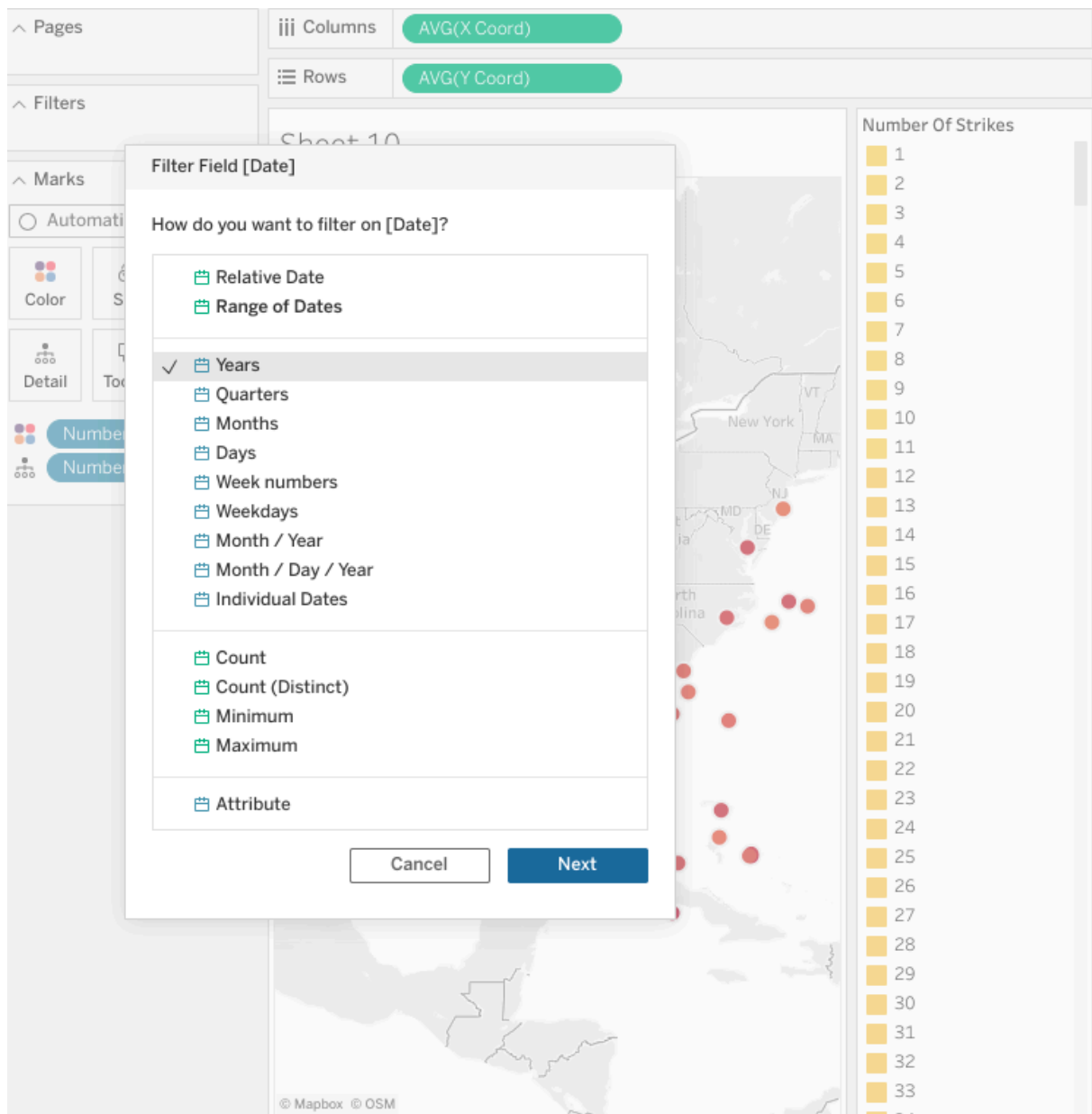
Note: Be sure to click the dropdown on "Number of Strikes" and convert it to a dimension.



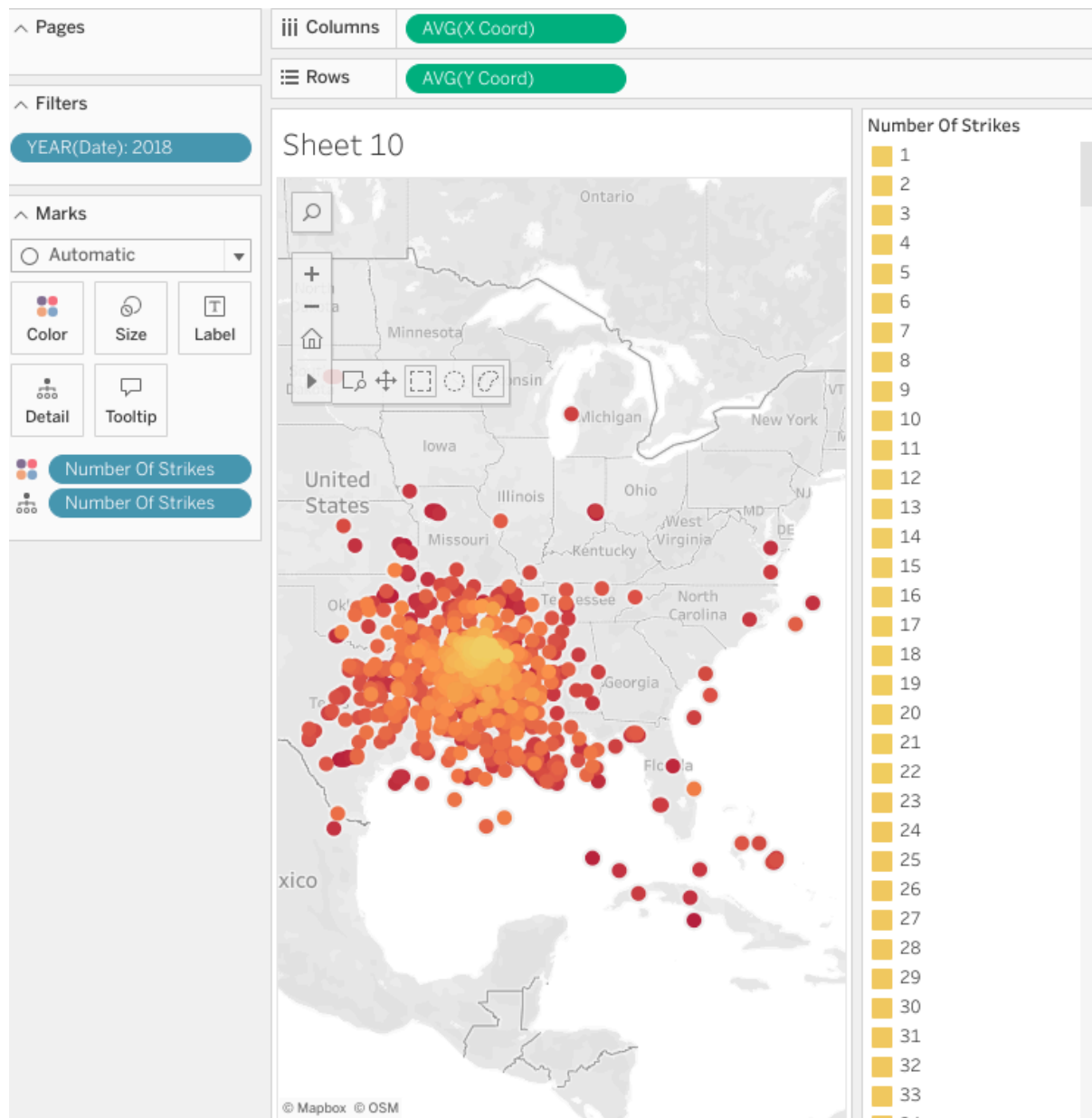
- Click on NEW WORKSHEET.
- Drag the X-coord to the column field. Drag the Y-coord to the Row field.
- Drag the Number of Strikes measure to the detail field.
- **Note:** Make sure the Number of Strikes measure is Discrete. To do so, right click Number of Strikes measure and click "Convert to Discrete."
- Add Number of Strikes to the color field and make the color coincide with the number of strikes. Select Red-Gold under "Automatic" drop down.



- Drop Date into the Filters field and select Year.



- Use the lasso tool. To do so, hover your cursor over your geographic map visualization and a small line of graphics will appear on the left side. Hover over the right arrow then click on the lasso image.



- Drag your cursor over the Texas state border, select Keep Only then click Create Set.
- Drag your new set to the Filters field and select 'In' from Filter.

Filter [In / Out of Number Of Strikes Set]

General

List: Select from list

Values: All values in data set

Search

☐ (All)

☒ In

☐ Out

☐ Exclude selected values

> Condition

> Top/Bottom

Summary

Selection: Selected 0 of 2 values

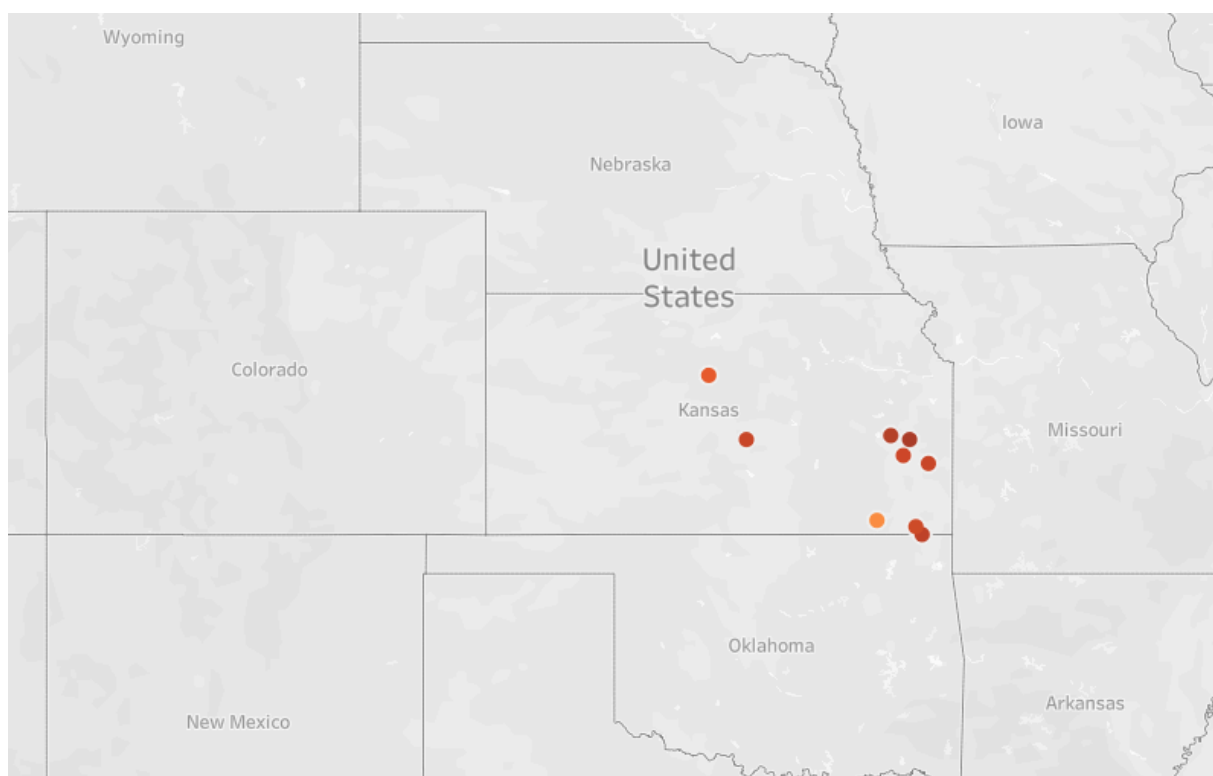
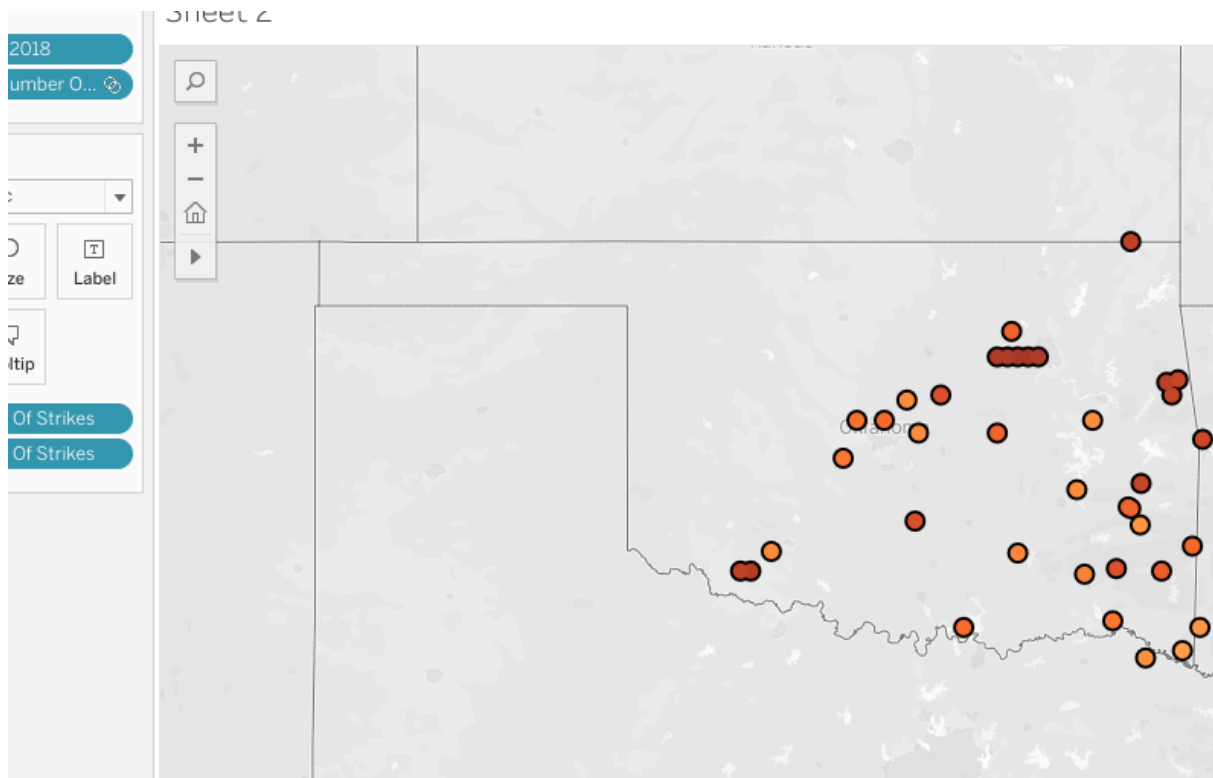
Wildcard: All

Condition: None

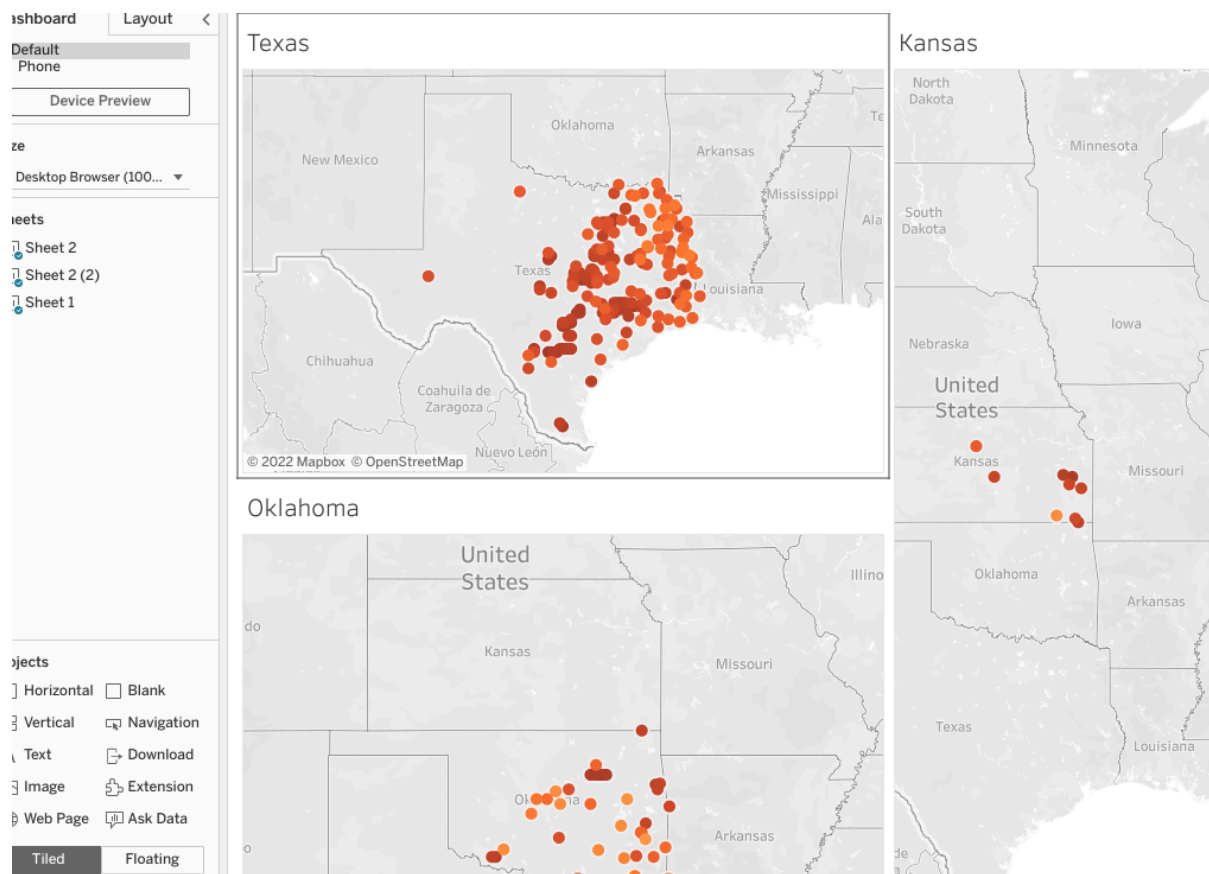
Limit: None

Reset Apply Cancel OK

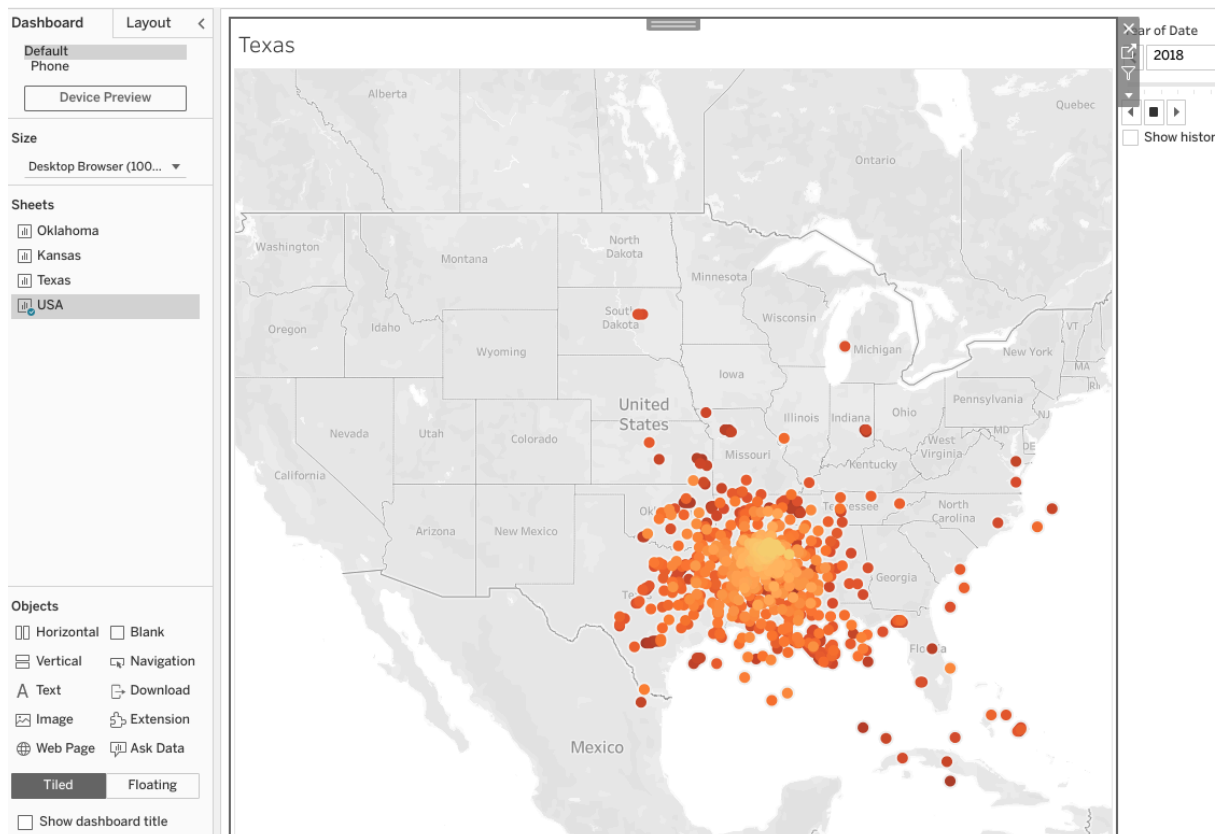
- **Note:** The dashboards section shown in the final part of this instructional video should be completed on your own. It is up to you to design the Oklahoma and Kansas visualizations by following the same steps as performed for Texas in the video. For detailed instructions to create the on-screen visuals shown in the remainder of this video, please follow the proceeding steps.
- Create two duplicates of this sheet and redo the same tracing of states process, but for Oklahoma and Kansas.



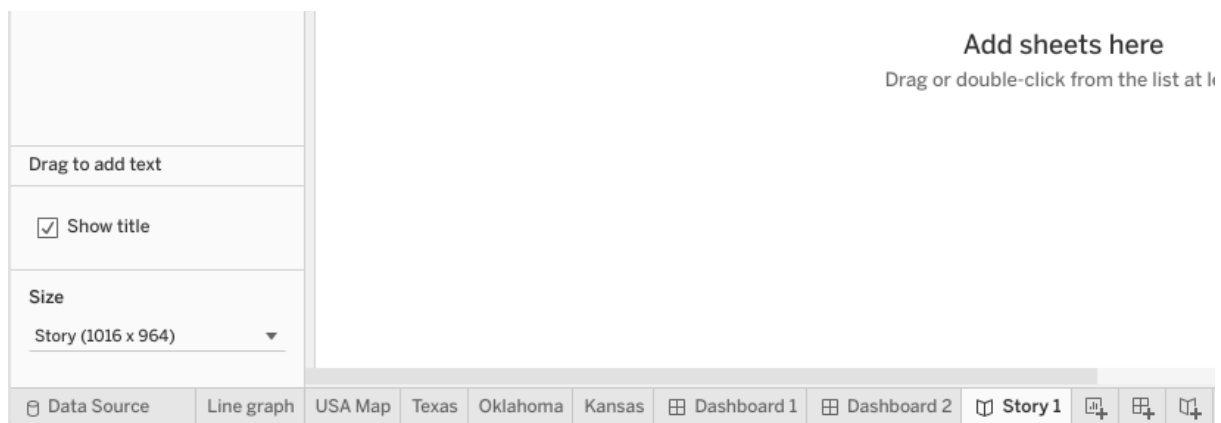
- Click on "New Dashboard" at bottom of page.
- In the dashboard, embed the three state worksheets.



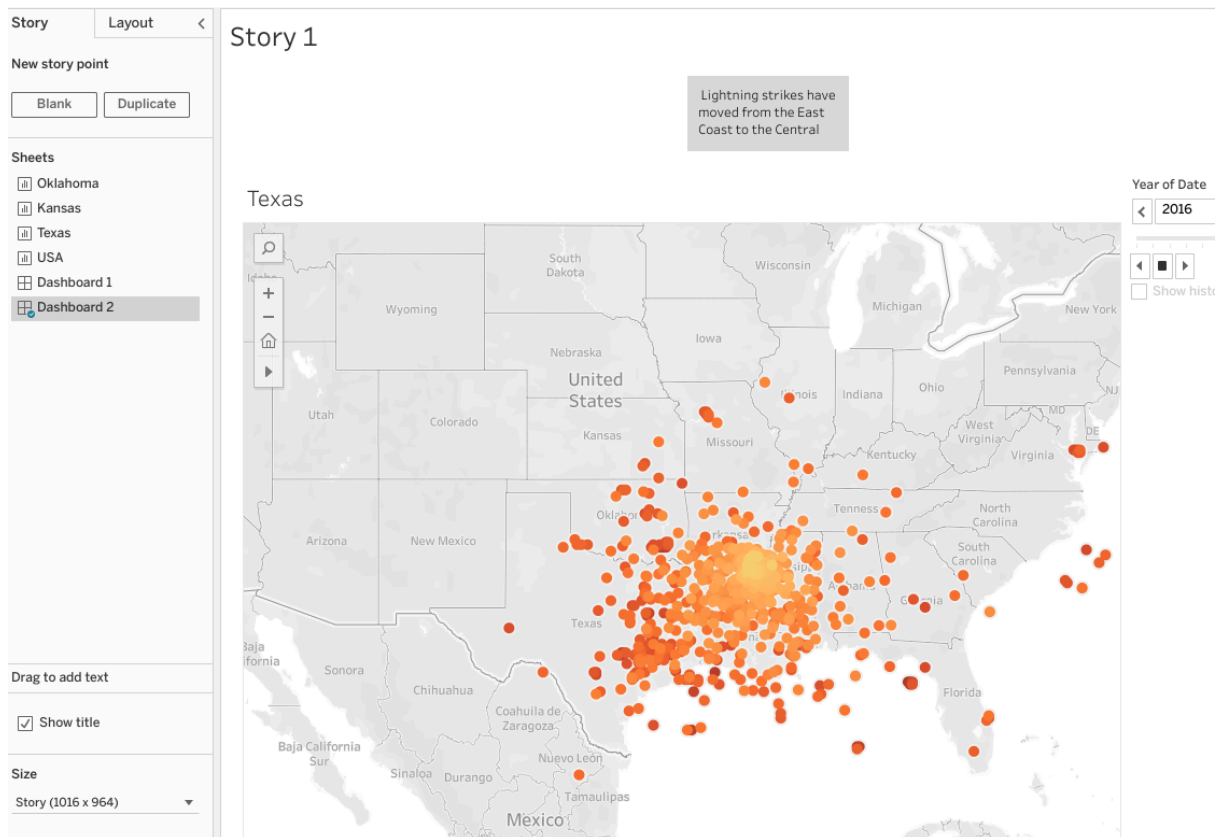
- Create another dashboard.
- Place USA map in it.



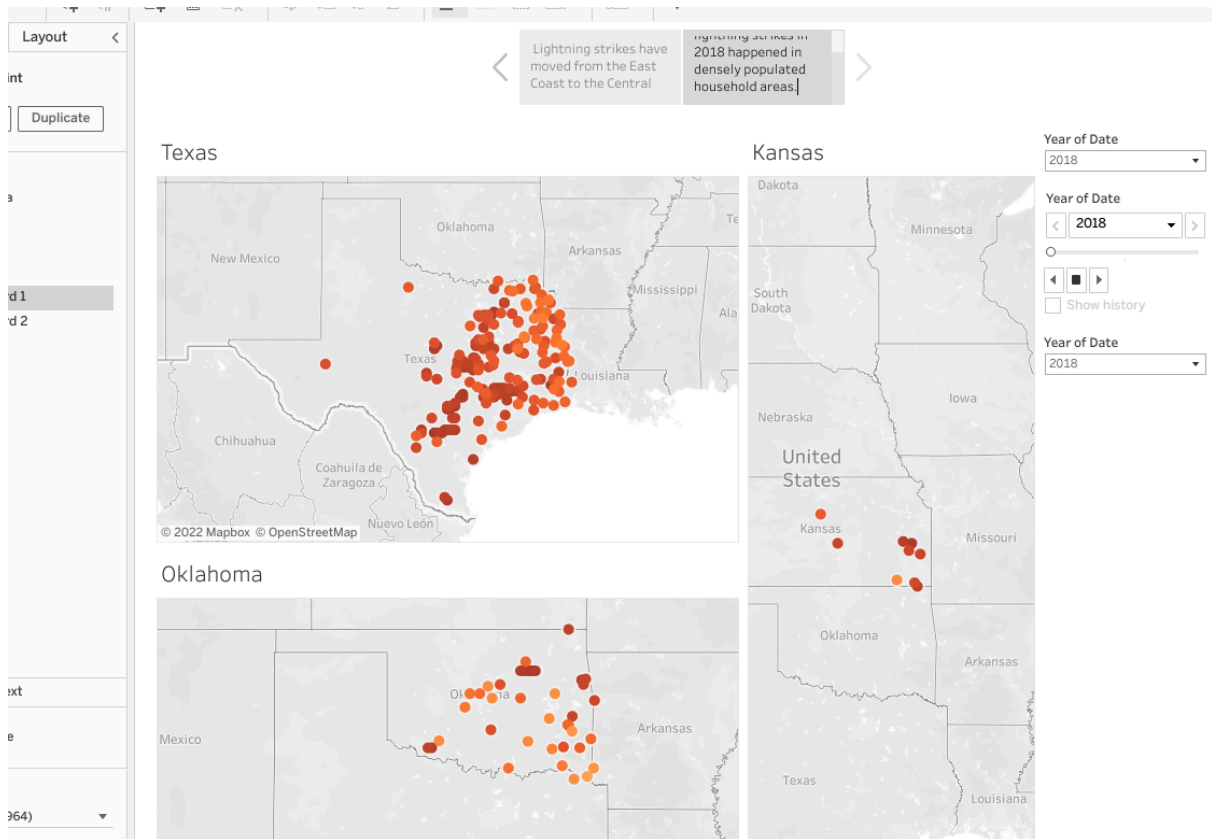
- Create a new STORY at the bottom of the Tableau page.



- On the first page, select the line graph dashboard.
- Fill in the caption with "Lightning strikes in the U.S. have increased 50% over the last decade."
- Go to page 2 of the story.
- Select USA dashboard for Page 2.
- In caption type, "Lightning strikes have moved from the East Coast to the Central Mainlands over the last decade."



- Go to page 3 of Story.
- Select the last dashboard with three states on it.
- In caption type, "Most number of lightning strikes in 2018 happened in densely populated household areas."
- And now you've created a story using dashboards and worksheets as building blocks!



The top five data visualization resources

Top five data visualization resources for data professionals

You've gotten a lot of experience in data visualization by now, which will be instrumental in your career as a data professional. As you step into new roles in the data industry, it is important to keep up with data visualization trends by participating in visualization ("viz") communities and accessing viz resources.

Below you'll find the top five data viz resources for data professionals. At each site you'll find amazing training resources and bustling data viz communities ready to help you maintain and improve on the skills you've already acquired. Along with improving your skills, these sites and communities can be a useful resource for maintaining a working knowledge of data viz ethics and accessibility.

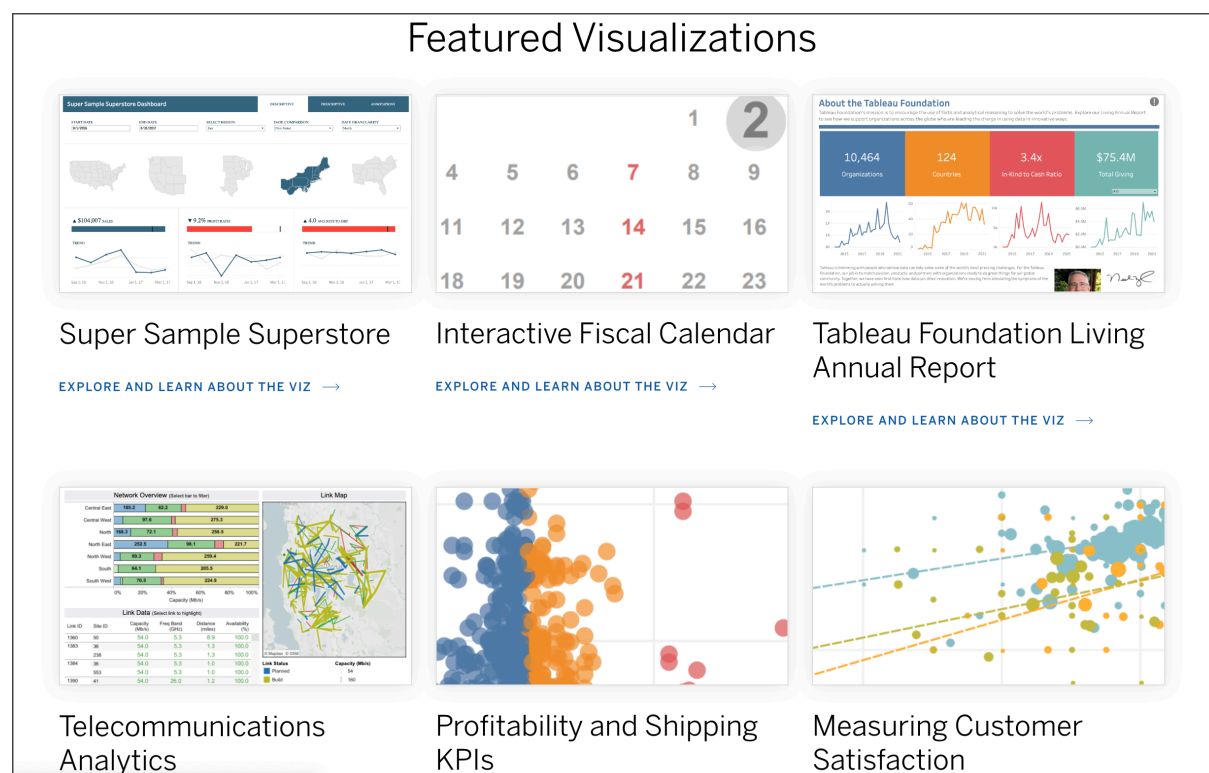
Take time to explore each resource. In particular, learn what they offer for training, education, and community engagement.

"Where can I find training material and sample visualizations to help me with my own data visualizations?"

Tableau

You're already quite familiar with Tableau's free browser software: **Tableau Public**. Of course, they offer a variety of software packages that appeal to every data professional, from the daily power user to the occasional presenter.

But did you know that Tableau also offers a comprehensive education and tutorial platform focused on training both software users and class leaders? In their **Resources** tab, Tableau offers video tutorials, e-learning courses, and webinars to help every level of user improve their data visualizations. Along with articles, blogs, and white papers, Tableau has a huge collection of sample visualizations (including **Viz of the Day**, which showcases some of the best Tableau vizzes) that is meant to encourage and empower data pros at every level to turn their dull datasets into effective storytelling tools.



"Where can I find a community of data professionals dedicated to building intelligent data visualizations for businesses? "

PowerBI

What **PowerBI** is probably best known for is its popular software product used by businesses and corporations worldwide for data visualization. Not only do they offer a free online version of their premium product, PowerBI, they also provide comprehensive **guided learning** and **online workshops** for teaching users of its products to create powerful data visualizations.

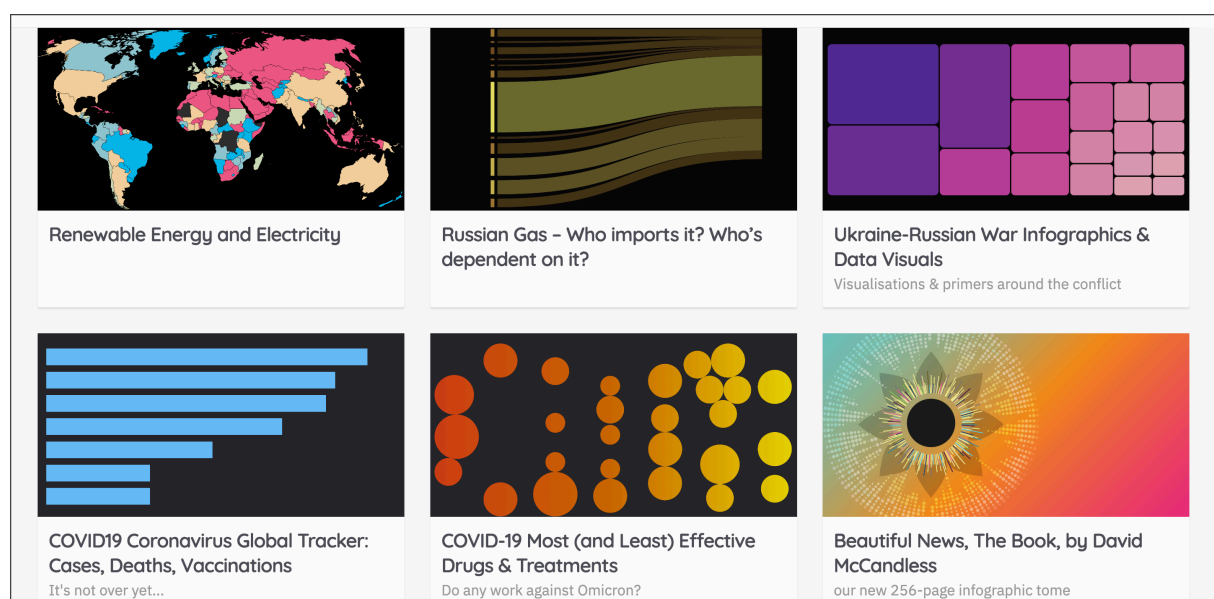
A growing number of data pros are also part of their **PowerBI data visualization community**. Here, users collaborate and share insight about the most effective ways to build data visualizations for businesses and organizations of every kind all over the world, like The Associated Press, Nokia, Real Madrid Football Club, Toyota, UNICEF, Galadari Brothers, and Eurobank, to name a few.

"Where can I find data visualizations paired with the datasets they are built from? "

Information is Beautiful

Founded by author and data visualization expert David McCandless, the [Information is Beautiful](#) platform is a gathering place for data pros who want to help people make clearer, more informed decisions in the world.

The site is a treasure trove of data visualizations on a host of topics. The Information is Beautiful team also has blogs, newsletters, and live and [online workshops](#), which aim to help data professionals improve their craft. Beyond the powerful visualizations and graphs on their website, Information is Beautiful focuses on transparency by sharing all of the datasets on which they base their vizzes.



"Where can I go to learn more about communicating my data visualizations effectively?"

Storytelling with Data

As evidenced by their name, [Storytelling with Data](#) is an organization focused on communicating with concise data that informs change. They provide training, workshops, and tools for the craft that combines both data visualization and storytelling.

Storytelling with Data has a particularly engaged [community](#), where data pros gather to learn, practice, ask for feedback, and help others. Though they don't have their own dedicated software for designing vizzes, they have a myriad of tiered resources for training, office hours, and consulting.



We help people and organizations
create graphs that make sense and
weave them into compelling, action-
inspiring stories

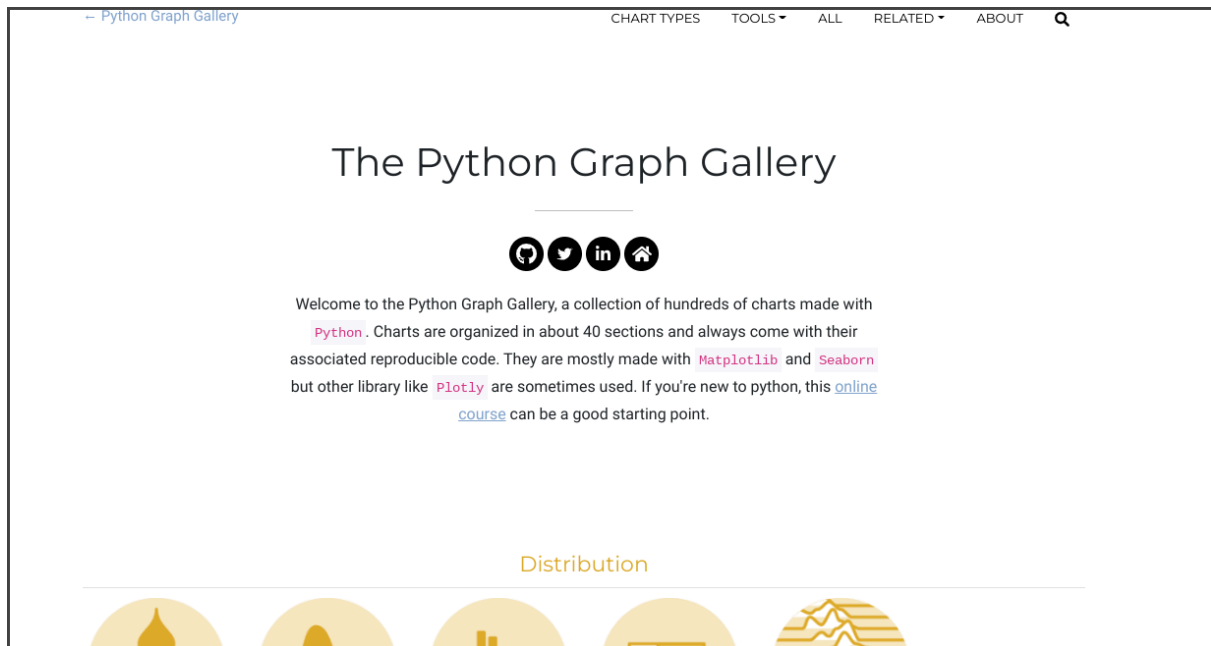


"Where can I go to find out more about data visualizations created in Python?"

Python Graph Gallery

If you are looking to improve your skill in plotting data visualizations in Python, the [Python Graph Gallery](#) is a fantastic website to visit. This site created by [Yan Holtz](#) includes a wide range of different types of visualization plots, like bar graphs, line charts, times series charts, geographic maps, and many more. Along with each visualization, you will also find the accompanying Python code used to plot the viz.

The site is organized first by plot type, but can also be sorted by the visualization Python package used, seaborn, matplotlib, or plotly. There is also a space for general knowledge tips and tricks, and a section titled [Dataviz Inspiration](#), which can help you understand what Python is capable of.



Key Takeaways

Your data visualization education doesn't end when you complete this program. There are amazing resources available that are consistently being updated with the latest trends and most up-to-date principles regarding the design, ethics, accessibility, and communication of data visualizations.

Follow-along guide: Present like a pro with Tableau

This document includes detailed instructions for how to perform the data visualizations described in the video "Present like a pro with Tableau."

The following guide points out areas of the video that may require adjustment. These resource guides can also serve as a set of usability reminders for you to recall when using Tableau in your future career.

Instructions

- Go to <https://public.tableau.com/s/>
- Since you've already [set up your Tableau Public profile](#), all you need to do is log in and select **Web Authoring** under **Create** in the navigation bar.
- Select the appropriate CSV file provided in the [instructions](#). Use the same data source as the previous video. The dataset you'll use with this instructional video is: `tableau_main_2009_to_2018.csv`
 - Please be aware that when you download the zip file folder provided, the computer automatically names that zip file folder with a long string of numbers and letters. You

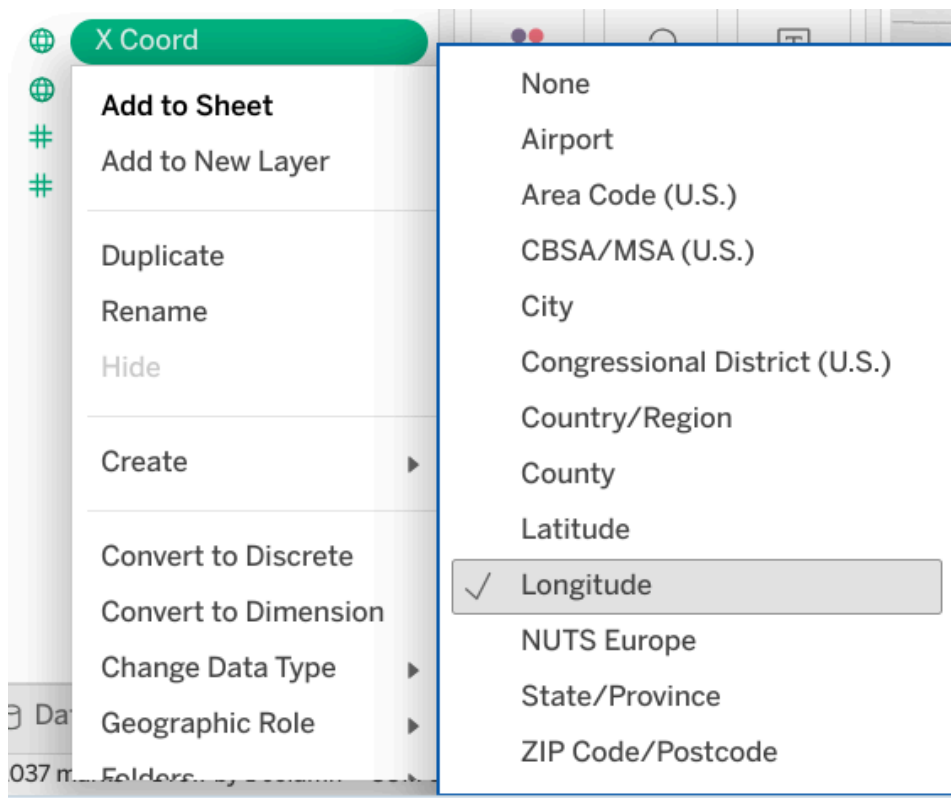
have to open that folder and then upload the individual files that are named correctly and match what's shown in the video

- Click on NEW WORKSHEET.

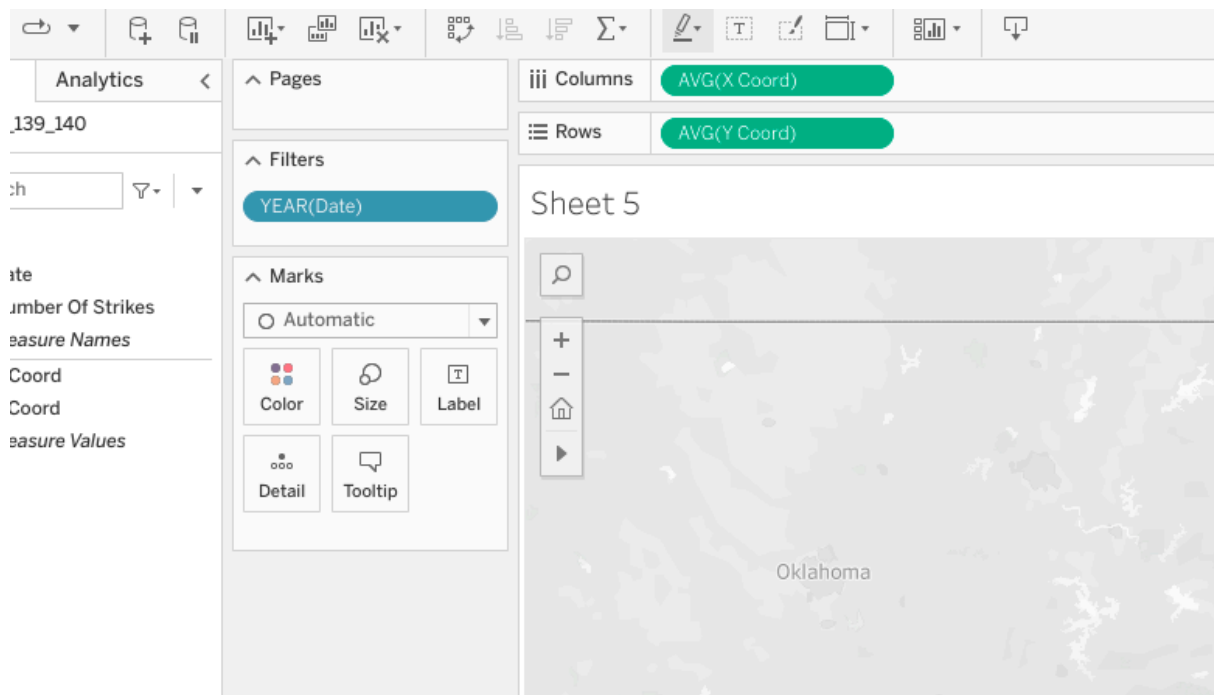
Note: Please allow several minutes for data import into a new worksheet.

Sheet 1- Map

- Set the geographic role of the X coord to Longitude
- Set the geographic role of the Y coord to Latitude



- Drag X coord to Column field. Drag Y coord to Row field.
- Drag DATE to FILTERS.



- Select **Years**. This will be a filter. Select All.

Filter [Year of Date]

General

List

Select from list

Search

☒ (All)

☒ 2009

☒ 2012

☒ 2015

☒ 2018

Values

All values in data set

☐ Filter to latest date value when workbook is opened

☐ Exclude selected values

Condition

Top/Bottom

Summary

Selection: Selected 0 of 4 values

Wildcard: All

Condition: None

Limit: None

Reset

Apply

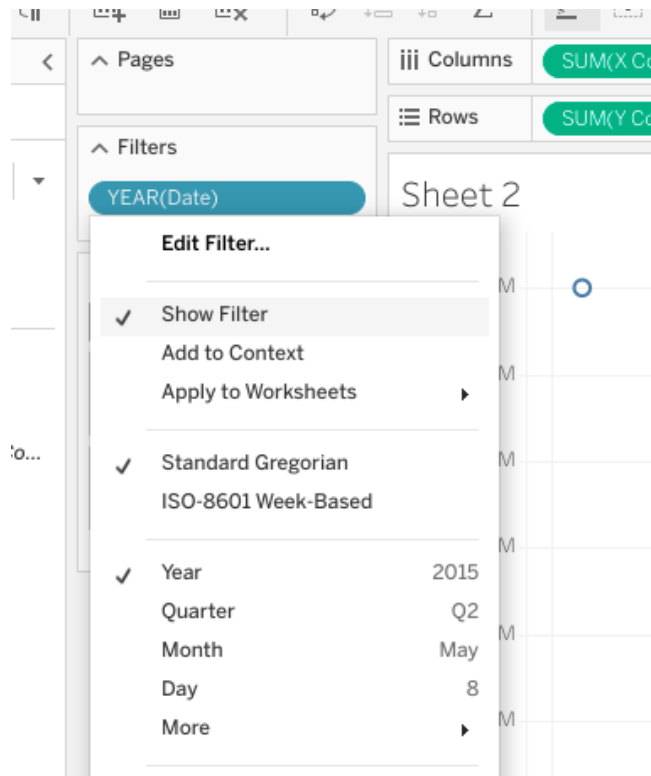
Cancel

OK

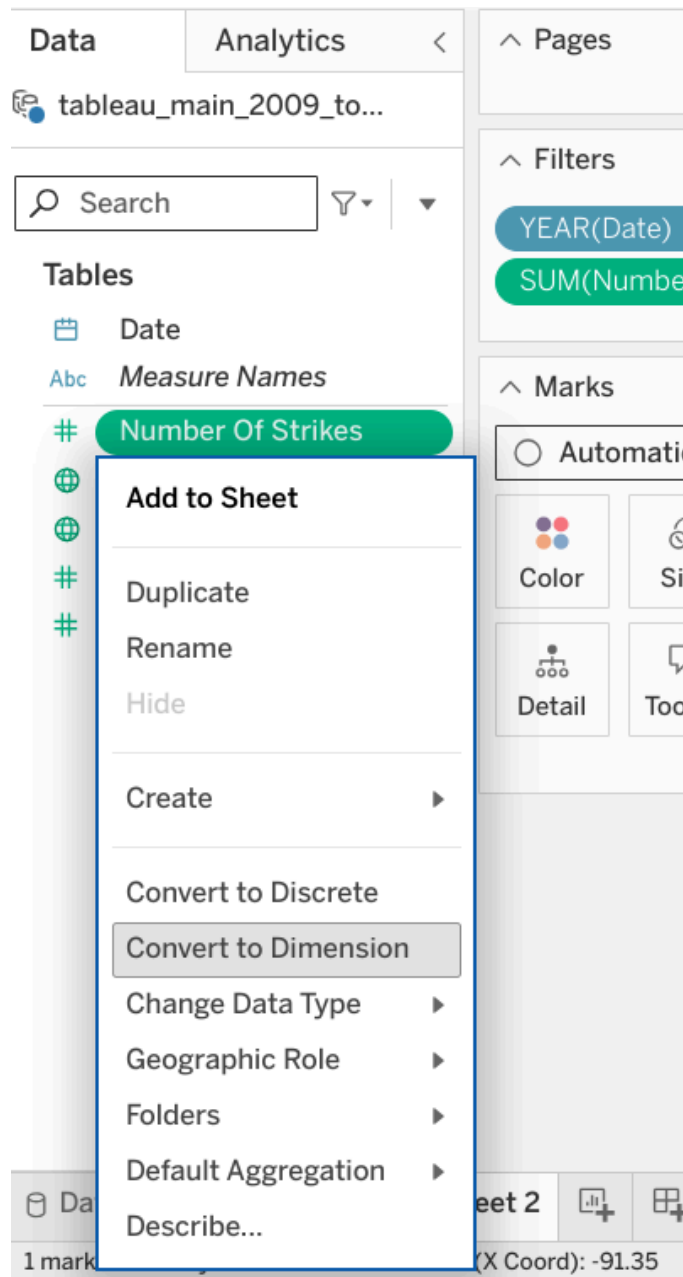
- Click on DATE dropdown, select SHOW FILTER.

Advanced

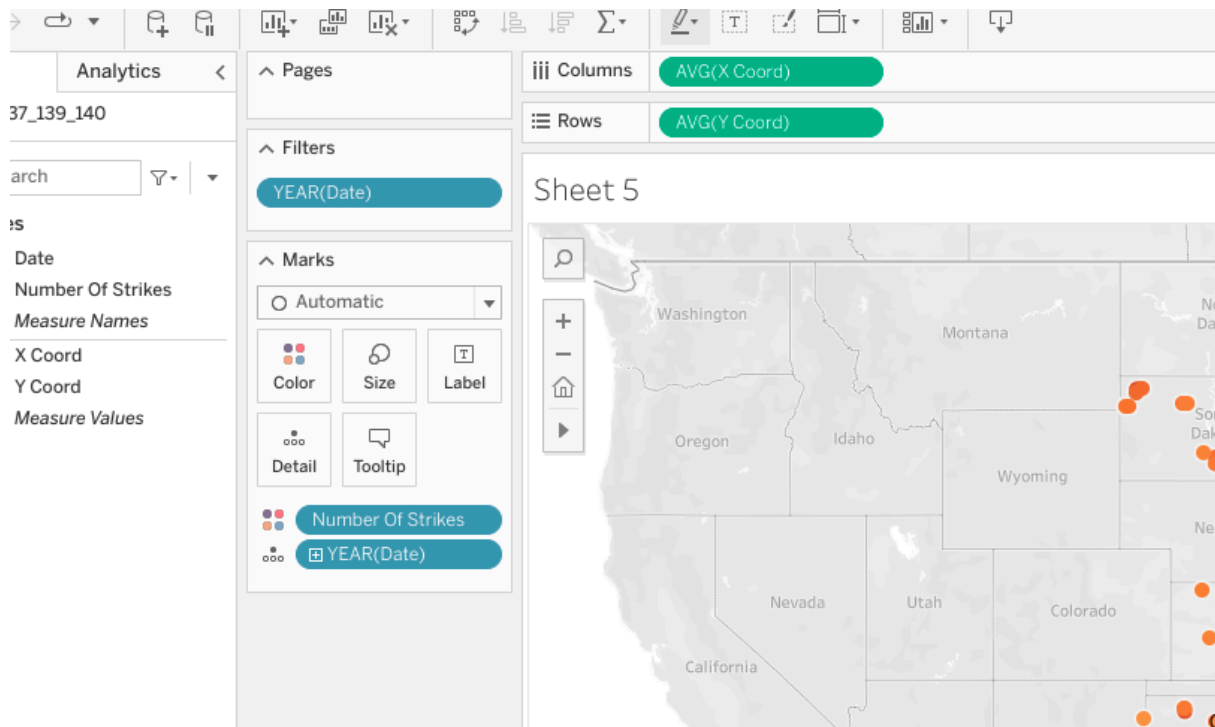
40



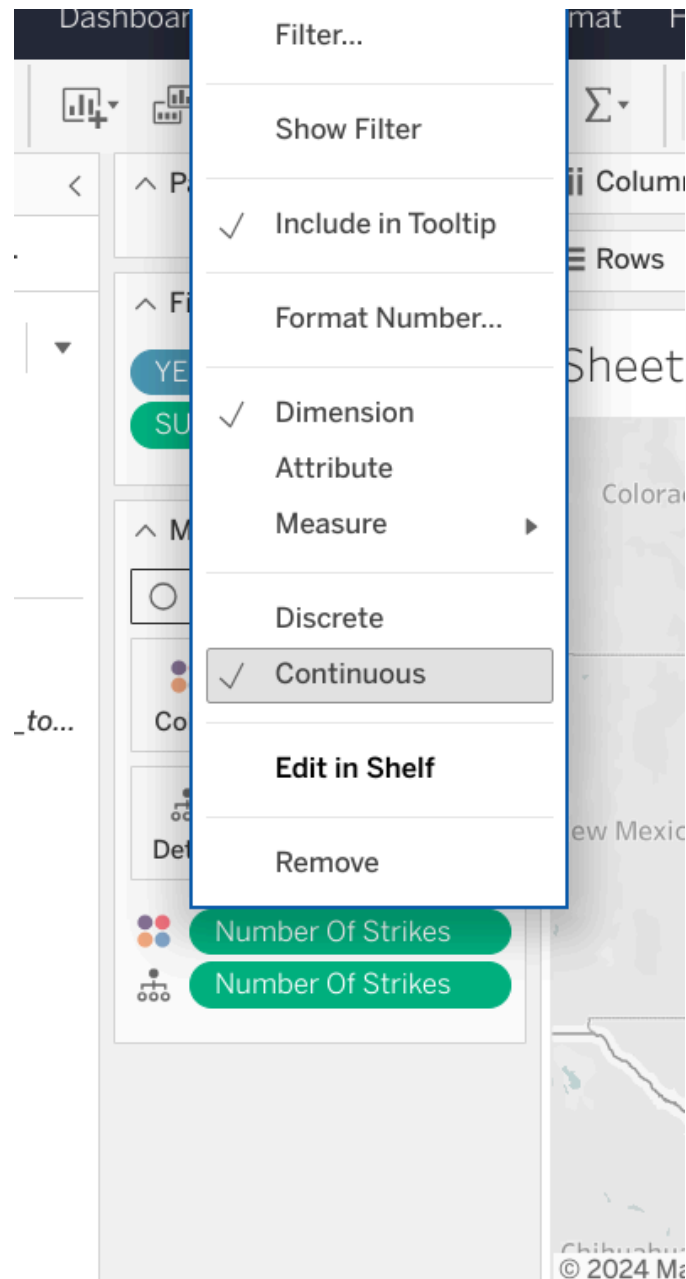
- Click on NUMBER of STRIKES and select CONVERT to DIMENSION



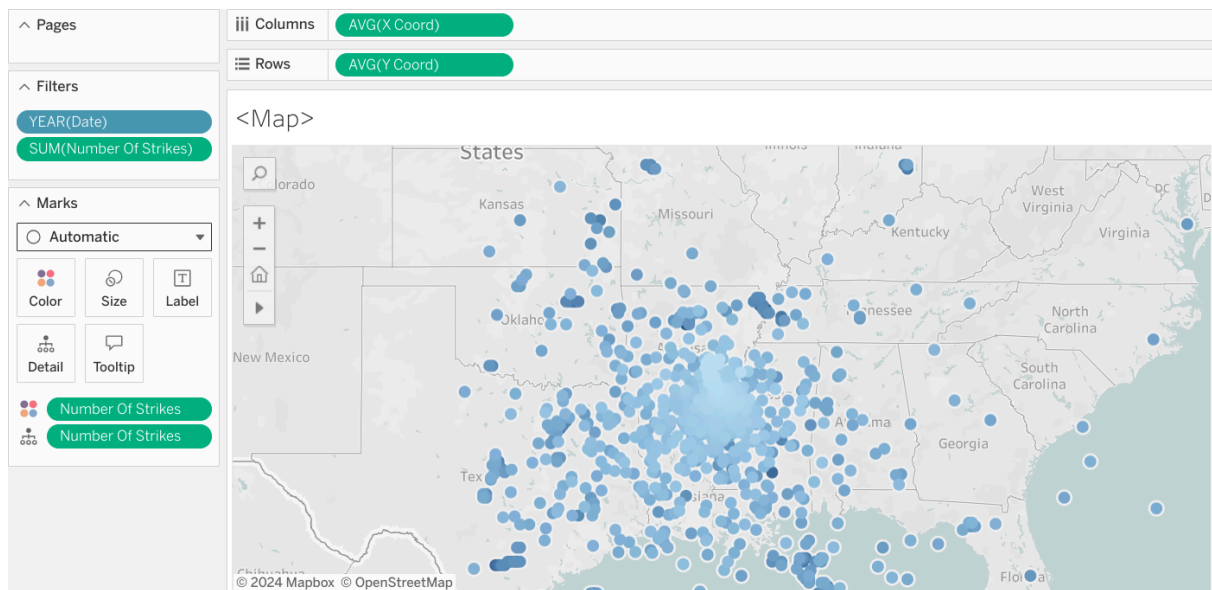
- Drag NUMBER of STRIKES into both the COLOR square and DETAIL square inside the MARKS box.



- Click on dropdown of NUMBER OF STRIKES, select CONTINUOUS.

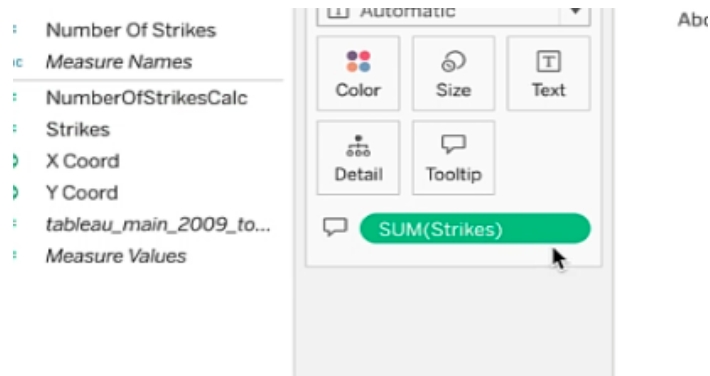


- Before moving on to the next Sheet, title Sheet 1 Map. You can do this under Rows between <> by clicking in that space.

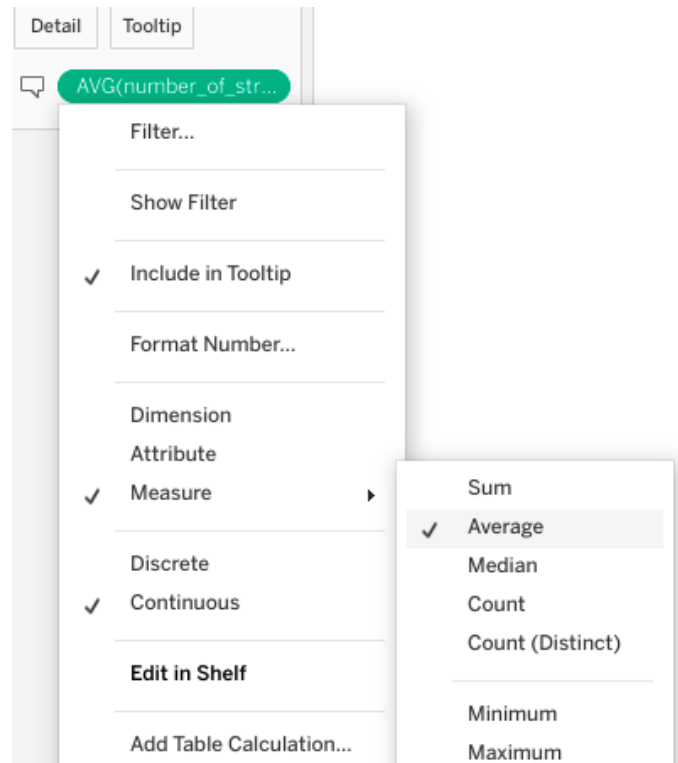


Sheet 2

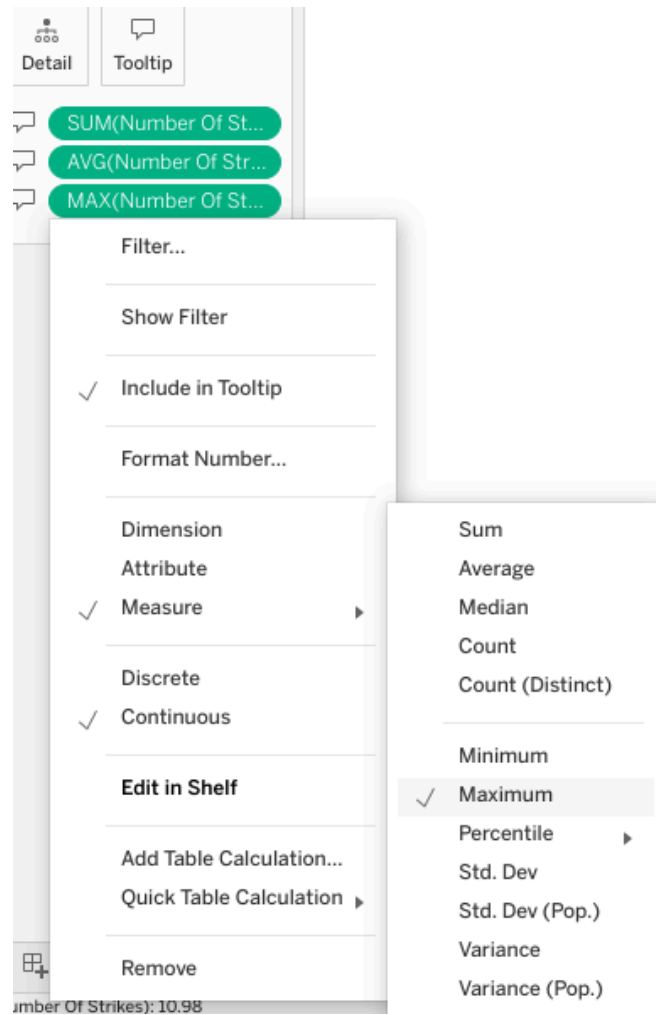
- In the same Tableau Workbook, Click on NEW WORKSHEET.
- Click dropdown of NUMBER of STRIKES field.
- Select CREATE and CALCULATED FIELD.
- TYPE "Strikes" for the title.
- Click OK.
- Drag calculation into TOOLTIP field.



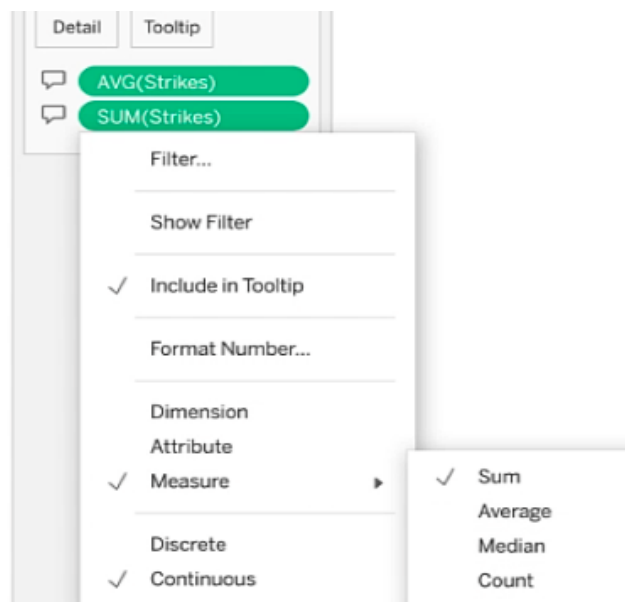
- In the Strikes drop down, Select MEASURE, then AVERAGE.



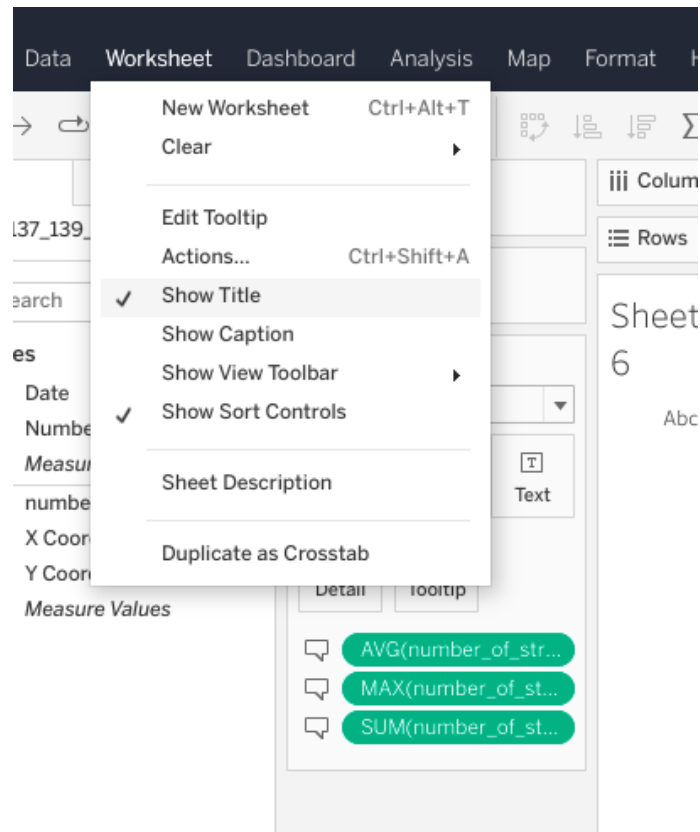
- Drag calculation field to TOOLTIP again. Select MEASURE, then MAXIMUM.



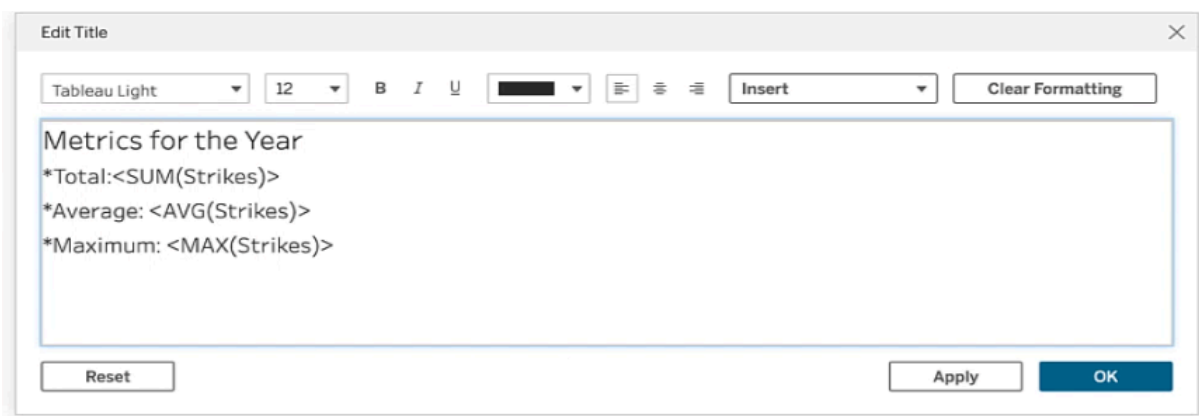
- Drag calculation to TOOLTIP, select MEASURE, then SUM.



- Click on WORKSHEET and select SHOW TITLE.

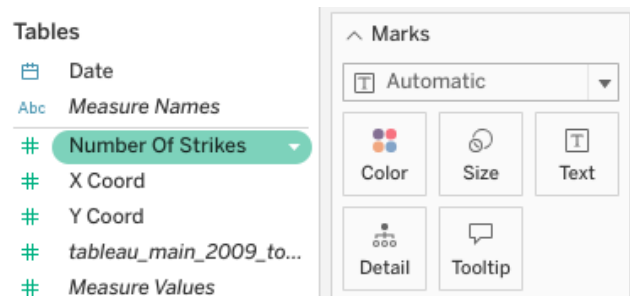


- Type in 'Metrics for the Year' in title field.
- Then add the following beneath it:
- Total: <SUM(Strike)>
- Average: <AVG(Strike)>
- Maximum: <MAX(Strike)>.
- Click OK.

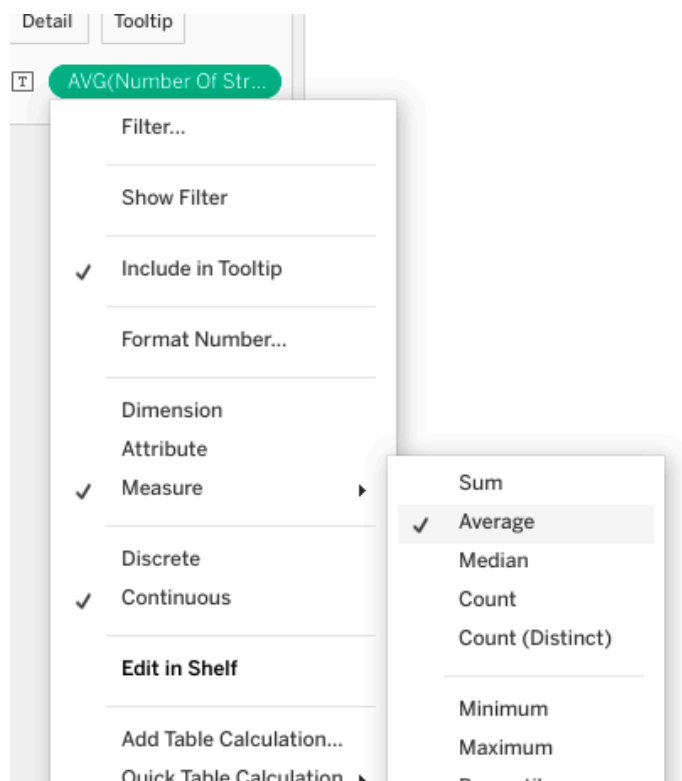


Sheet 3

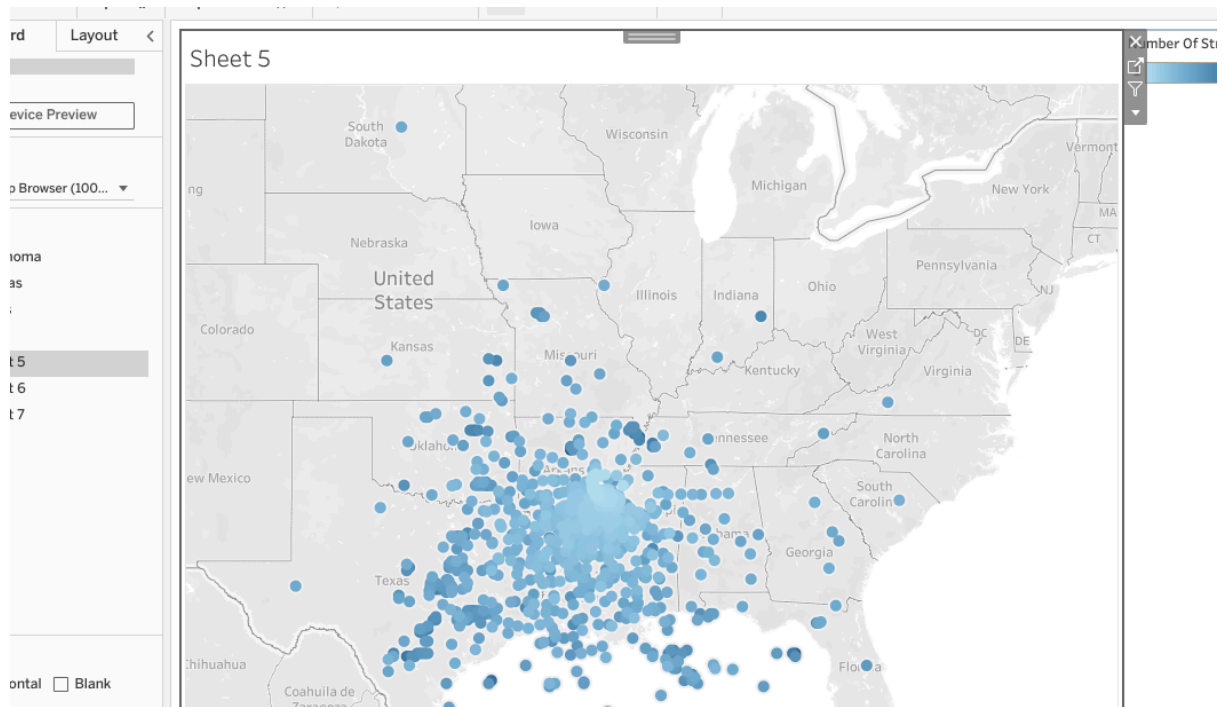
- Click on NEW WORKSHEET.
- Drag NUMBER of STRIKES to the TEXT square in the MARKS field.



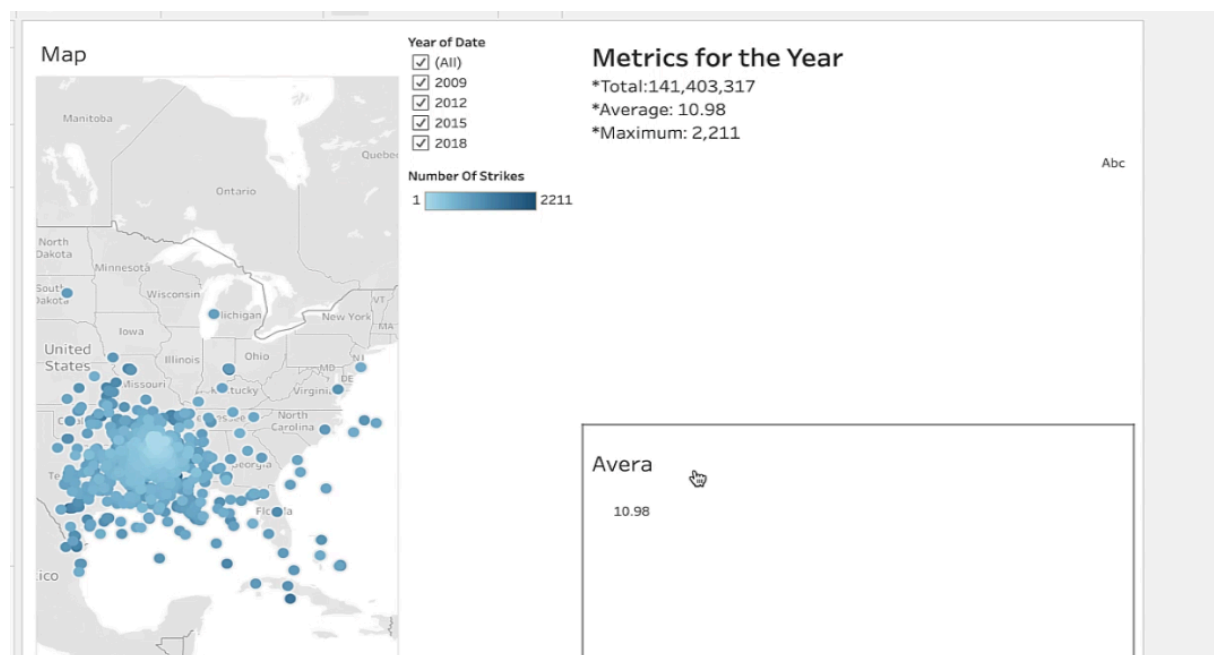
- From dropdown, select MEASURE and AVERAGE.



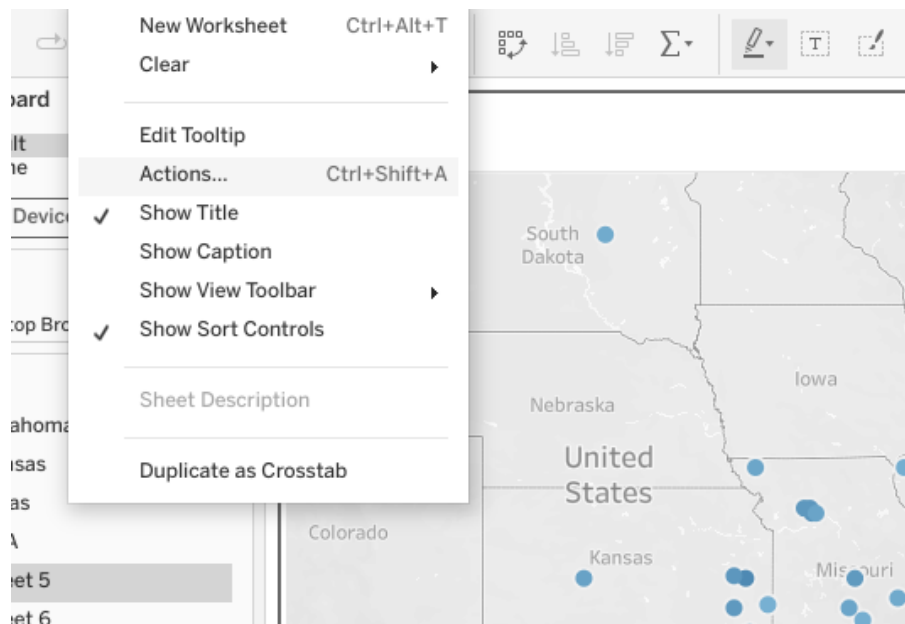
- Click on NEW DASHBOARD.
- Drag USA map worksheet to empty dashboard.



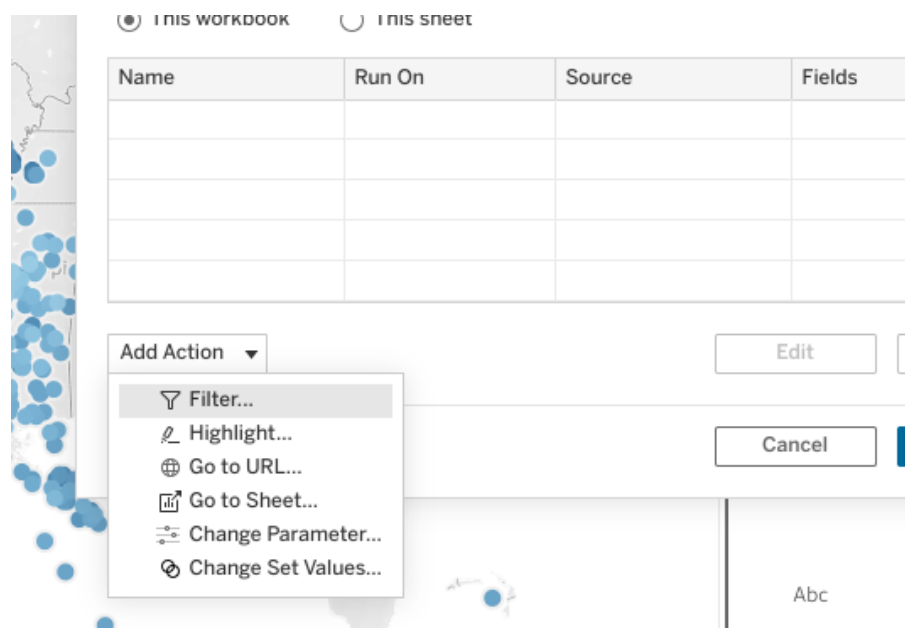
- Drag your other two worksheets to the dashboard.



- Click map, then select WORKSHEET and ACTIONS.



- Select ADD ACTION and FILTER.



- Under SOURCE SHEETS, select the dashboard you created from the dropdown.
- Check the box for the map worksheet, and then choose SELECT under the RUN ACTION ON list.

Source Sheets

Dashboard 3

☒ MAP
☐ Sheet 6
☐ Sheet 7

Run action on

☐ Hover
☒ Select
☐ Menu
☐ Single-select only

Target Sheets

MAP

Clearing the selection will

☒ Keep filtered values
☐ Show all values
☐ Exclude all values

Filter

- Next, under the TARGET SHEETS list, select dashboard from the drop down, then check box for the other two worksheets you created.
- Select SHOW ALL VALUES.
- Select ALL FIELDS, under Filter.
- Click OK.

Target Sheets

Dashboard 3

☐ MAP
☒ Sheet 6
☒ Sheet 7

Clearing the selection will

☐ Keep filtered values
☒ Show all values
☐ Exclude all values

Filter

☒ All fields ☐ Selected fields

☐	Source Field	Target Data Source	Target Field
☐	Click to add		

- Type in 'Interactive Filter' in Legend title field.
- Type in 'Location Metrics' in title field.
- Now you've created an interactive geographic dashboard that adjusts metrics each time a year or location is changed or selected. Well done!

Activity Exemplar: Build an interactive dashboard in Tableau Public

Here is a completed exemplar along with an explanation of how the exemplar fulfills the expectations for the activity.

Completed Exemplar

To review the exemplar for this course item, click the following link .

Link to exemplar: [Impact of Holidays on Seoul Bike Rentals in 2017 - 2018](#)

Assessment of Exemplar

Compare the exemplar to your completed activity. Review your work using each of the criteria in the exemplar. What did you do well? Where can you improve? Use your answers to these questions to guide you as you continue to progress through the course.

Note: *The exemplar represents one possible way to complete the data visualization. Yours will likely differ in certain ways. What's important is that your data visualization meets the business scenario criteria.*



How the exemplar meets the criteria

Notice how the exemplar posted to Tableau Public uses the detail from the given scenario to adhere to the following guidelines:

- The data dashboard includes two dynamic worksheets.
- The dashboard has divided the data into two different bar graphs: by Weekday and by Adjacent Day.
- One worksheet divides data into dates and months
- This worksheet is the "By Adjacent Day" worksheet.
- Another worksheet is divided into dates for each day of the week. This worksheet is the "By Weekday" worksheet.
- The dashboard, worksheets, and filters are properly labeled.
- Holidays are shown in a clearly contrasting color. Holiday data is in orange, while non-holiday data is in blue.
- The filters on the dashboard sidebar filter the desired data.
- There are two filters shown and one legend.
- One filter connects to the "By Adjacent Day" data and allows the data to be filtered by month.

- Another filter connects to the "By Weekday" data and allows the data to filter by day of the week.
- The legend shows that holidays are orange and non-holidays are blue.